OPSC6312 OPEN SOURCE CODING (INTERMEDIATE) · PORTFOLIO OF EVIDENCE

Nexus: Part 3 Final Submission

Final asset development, additions, tweaking and final submission (Learning Units 1–4). App version 2.0.0.

Assessment sheet details

Module	OPEN SOURCE CODING (INTERMEDIATE) · OPSC6312
Students	Clarity Masuku (ST10438928) · Sibusiso Mabena (ST10462532). Names and numbers as listed in the repository README history; **confirm before submitting**.
App	Nexus 2.0.0 (versionCode 2), package `com.example.nexus`
Repository	https://github.com/jBlack-MC/Nexus-app (Kotlin source on GitHub; no ZIP)
Final commit / tag	Team action Commit the Part 3 working tree, then tag it: `git tag -a final-poe -m "Final PoE"` and `git push origin final-poe`
Demo video	Team action Unlisted YouTube link (recorded on a physical phone; script in section 5)
Date	2026-11-05
Document status	Prepared with AI assistance (see section 6). Every statement is checked against the repository and test runs. The team must review it and fill in the marked team actions.

Labels: Implemented verified in code or test output · Inferred · Recommendation · Open question / team action · Unknown · Estimate

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1 Submission checklist

This table maps every item the brief says to "present for your submission" to where it is and what remains.

Part 3 submission items

Item required by the brief	Where	Status
Complete Kotlin source on GitHub (no ZIP)	`app/`, `backend/` on `jBlack-MC/Nexus-app`	Implemented Part 3 changes are **uncommitted**. Team action commit, push and merge to `main`.
Comments in code, with references	KDoc on all new classes (e.g. `GoogleSignIn.kt`, `Reminders.kt`, `Push.kt`, `push.js`, SSO block in `server.js`) with "Reference: … (accessed 2026-11-05)" lines; README references section	Implemented
Logging that shows understanding	Android `Log.i/w` in auth, SSO, push, reminders and notifications; Pino structured logs with request IDs and redaction in the API (e.g. "Google sign-in", "Push notification sent", "Refusing to start")	Implemented
README on GitHub: purpose, design, GitHub and Actions use, images	`README.md` (rewritten for v2.0, 8 real screenshots)	Implemented
Video presentation with voice-over, link in README	Script: section 5	Team action record on a physical phone; add the link to the README and this cover
Release notes since the prototype, naming the innovative features	Section 3; `README.md` → Release notes	Implemented
Signed APK	Signing config, upload key, verification in section 7	Implemented pipeline verified. Team action build the final AAB/APK once the API has an HTTPS URL.
Screenshots	`docs/screenshots/part3/` (21), Play-ready set in `docs/play-store/assets/phone/` (8)	Implemented emulator captures. Team action the brief wants the *app running on a phone*: add phone photos or screenshots in the video.
Play Console screenshot (if applicable)	`docs/play-store/README.md`	Team action requires a Google Play developer account
AI-use write-up (≤ 500 words)	Section 6	Implemented draft; the team confirms and signs
Final commit tagged as the final PoE	Cover page	Team action

2 Requirements and rubric evidence

For each criterion this table gives the evidence and what limits it. It is not a mark: the lecturer assesses against the rubric. Requirement IDs link to the product requirements in the appendix.

Final PoE and general rubric criteria

Criterion	Evidence	Gaps / what reaches the top band
Feature: SSO login [10]	Google sign-in via Credential Manager; server verifies the ID token with Google; creates or links accounts (FR-21). 3 backend tests, 2 app tests.	Needs a Google Cloud Web and Android OAuth client to run for real; show it working on the phone in the video.
Feature: Offline mode [10]	Room outbox, sync banner, conflict review, idempotent replay (FR-16, FR-17). Verified on device: offline create, then reconnect, then the server has it (screenshots 19–20).	A blocked queue needs a manual "Discard" (AUD-19).
Feature: Real-time notification [10]	FCM push of cross-device changes, test push, plus WorkManager due-date reminders (FR-22, FR-23). Reminder and local notification verified on device (screenshot 13); 5 backend push tests.	Needs a Firebase project (`google-services.json` , service account) to show a real push from the server in the video.
Feature: Multi-language [10]	English, isiZulu, Setswana on every screen, 210 strings each (FR-18); verified on device (screenshots 14–18).	Translations are AI-drafted: have first-language speakers review them.
Feature: User Defined 4 [10]	Confirm against Part 1 Suggested: **habit tracking with gamification**: streaks, points, levels, badges, rewards, CSV export (FR-13 to FR-15).	The Part 2 PDF does not name the user-defined features; use the names from your Part 1 design.
Feature: User Defined 5 [10]	Confirm against Part 1 Suggested: **Today planning**: overdue, due-today and upcoming schedule, with task priority, status, labels, checklists and filters (FR-09, FR-11, FR-12).	The Part 2 PDF also claimed "list/board task views" and "sorting"; no board view or sort control exists in the code. Do not claim them.
Prepared for Play Store [5]	Signing config and upload key; signed release verified; Play kit with icon, feature graphic, 8 screenshots, listing, Data safety, privacy policy (section 7).	Upload to Play Console (Internal testing) and screenshot it; host the privacy policy.
Runs on a mobile device [5]	Verified on an Android 7.0 (API 24, minSdk) emulator; no crashes after the Settings fix.	The brief requires a **physical phone** in the video.
Register / login with encrypted password, settings, REST API [requirements]	bcrypt cost 12; JWT; Settings screen; Express API with 29 tests (FR-01 to FR-08).	The API must be **hosted** ("exists on the internet") and the video must show the stored data.
AI-use write-up [5]	Section 6: tools, what each did, how output was checked, citations.	The team adds anything else it used.
User interface [10]	Material 3, consistent spacing, colour and type tokens, bottom navigation, skeletons, clear errors, confirmation before every delete, labels that fit in all three languages (NX-04-DES).	No accessibility audit recorded (IMP-11).

Criterion	Evidence	Gaps / what reaches the top band
GitHub, README, automated testing [10]	Many commits and PRs; README; Android CI (ktlint gate, unit tests, APK) and Backend CI (tests, Docker smoke test) workflows; 29 + 42 + 16 tests pass locally; 0 lint violations.	**CI runs have not started since 2026-09-26 (account billing lock).** Fix it so the final commit shows green checks (AUD-04).
Demonstration video [5]	Script in section 5.	Record it: professional, voice-over, all features, phone, data in the API and authentication service.

3 Release notes: 2.0.0 (changes since the Part 2 prototype)

Innovative features added in this version

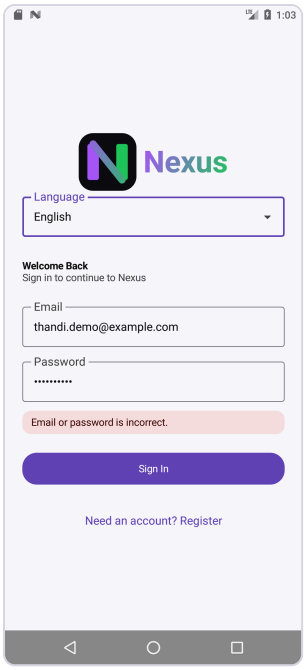
- **Google single sign-on** (FR-21): Credential Manager plus server-side ID-token verification; account linking by email.
- **Real-time push notifications** (FR-22): changes on one device push to your other devices, which refresh their offline copy; Settings has a test push.
- **Due-date reminders** (FR-23): WorkManager job, works offline and when the app is closed.
- **Full isiZulu and Setswana** (FR-18): every screen, error, sync message and notification.
- **Offline-first editing with sync** (FR-16, FR-17): added on 2026-10-04 after the prototype; Room outbox, sync banner, duplicate-proof replay, conflict review.

Improvements and fixes

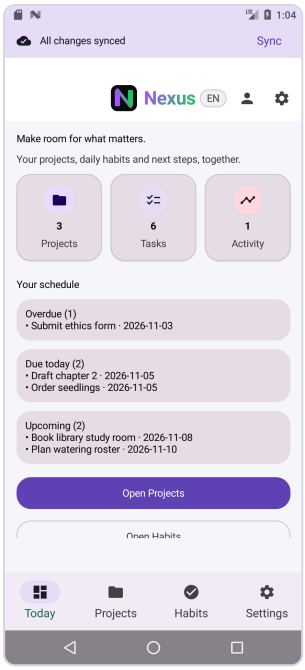
- Sign-in and registration explain wrong credentials, a taken email and rate limiting (AUD-18); emails are case-insensitive; sign-in timing no longer reveals registered emails.
- The server refuses to start on a damaged data file and keeps a `.bak` copy (AUD-01); compatible with Zod 4 (AUD-03); `npm audit` reports 0 vulnerabilities; proxy-aware rate limiting (AUD-06); root Docker Compose fixed (AUD-02).
- Fixed a crash when opening Settings; correct CSV quoting (AUD-11); correct multi-habit completion rate (AUD-17); offline cache excluded from cloud backup (AUD-05).
- Delete confirmations for projects, tasks and habits (IMP-12); single-line bottom-bar labels in every language.
- Codebase at 0 ktlint violations, with `ktlintCheck` gated in CI; dead code, the placeholder workflow and a stale failing build log removed (AUD-13, AUD-15).
- Release signing configuration and upload-key script; Play Store kit; version 2.0.0; CI on Node 22.

4 Screen evidence

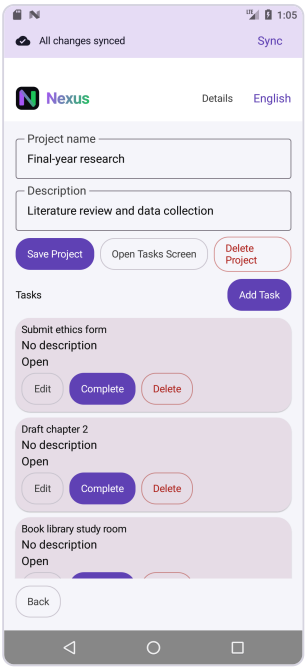
All screenshots come from version 2.0.0 running on an Android 7.0 emulator against the local API on 2026-11-05, using a fictional demo account. The brief asks for the final app on a physical phone: use these as supporting evidence and show the phone in the video.



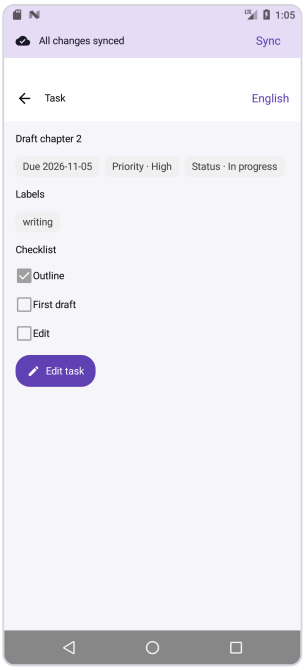
Clear sign-in error



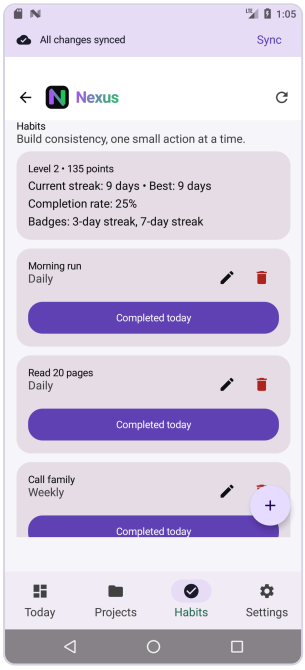
Today



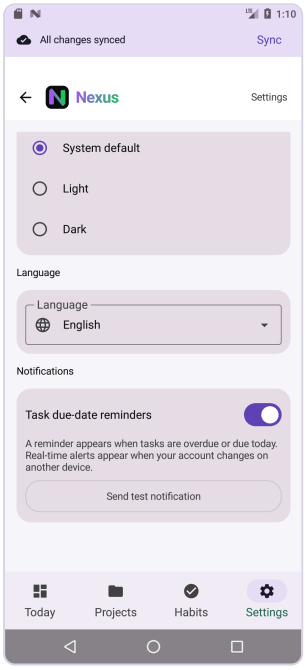
Project detail



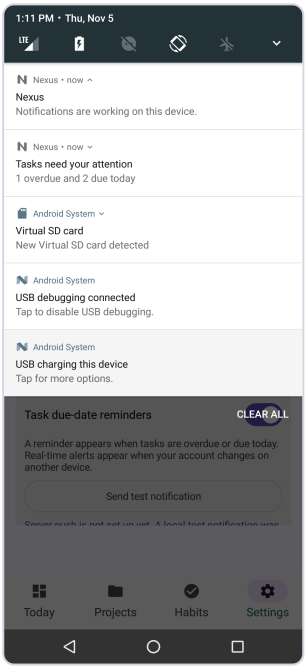
Task detail



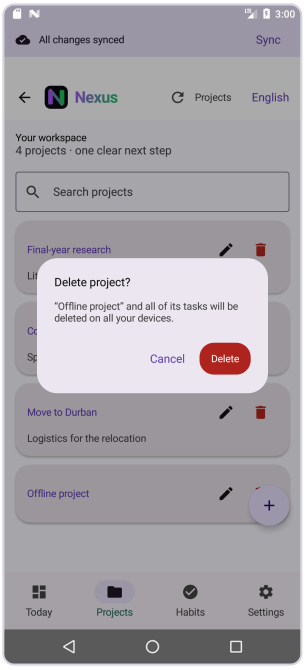
Habits



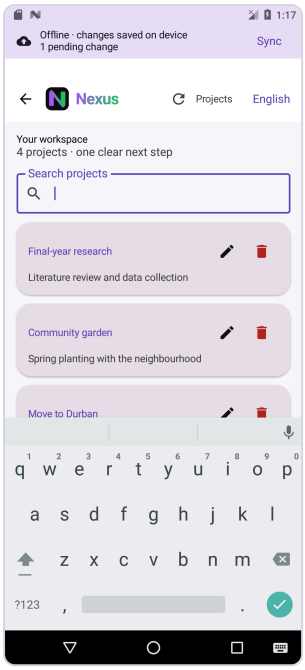
Settings: notifications



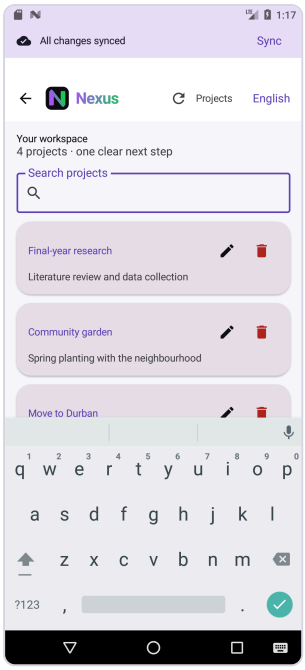
Reminder + test notification



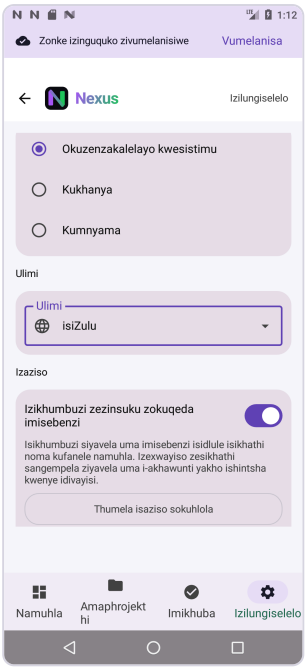
Delete confirmation



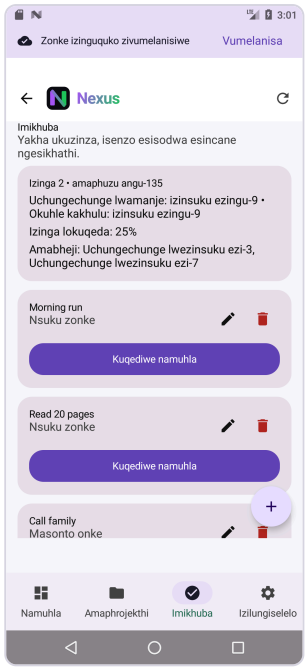
Offline: 1 pending



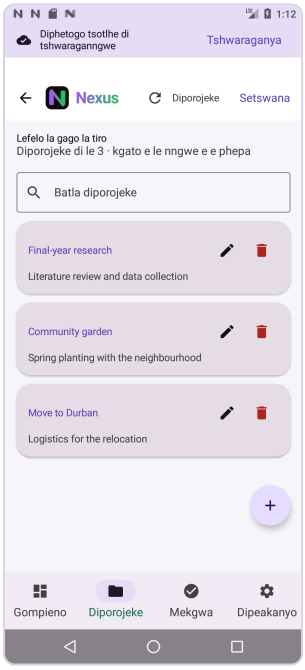
Reconnected: synced



isiZulu



isiZulu habits



Setswana

5 Demonstration video script (about 8 minutes, physical phone, voice-over)

Before recording: deploy the API over HTTPS, configure Google SSO and Firebase (NX-11-DEP), and install the signed build on the phone. Use a demo account, never real passwords, and do not show tokens or secrets on screen.

Recording plan mapped to the rubric

Time	Show	Say (key points)
0:00	Phone home screen, app icon, launch, splash	Nexus 2.0, what it is for, that this is a physical phone.

Time	Show	Say (key points)
0:30	Register a new user; wrong password, then the correct one	Validation; passwords hashed with bcrypt on the server; JWT stored encrypted on the phone.
1:30	"Continue with Google"	SSO: Google returns an ID token, which our API verifies before issuing a session.
2:15	API data: the hosted `/health` and the users in the data store (hashes visible, not passwords), plus the Firebase console's registered app	"What data is stored in the online-hosted authentication service, API and database."
3:00	Create a project and tasks with due date, priority, labels and checklist; filters; Today cards	User-defined features (Part 1 names).
4:00	Habits: complete today, streaks, badges; CSV export	Gamification.
4:45	Airplane mode on → create and edit → banner shows pending → airplane mode off → synced → item visible in the API data	Offline mode with sync and duplicate-proof replay.
5:45	Second device or emulator signed in to the same account: edit there, and the phone receives a push. Then Settings → Send test notification, and a due-date reminder	Real-time notifications (FCM) and scheduled reminders.
6:45	Switch to isiZulu, then Setswana, on Login and Settings	Multi-language support for two South African languages.
7:15	Settings: theme, password, delete account	Settings menu.
7:40	GitHub: README, Actions green checks, tag `final-poe`; Play Console Internal testing	Automated tests, CI and publication preparation.

6 AI-use write-up

Tools used. OpenAI Codex (September 2026; cited in Part 2) helped draft Kotlin implementation ideas and explain build errors. Cline (VS Code extension) and the GitHub Copilot coding agent were used during September 2026; their checkpoints and commits are visible in the Git history. Dependabot proposed dependency updates. For Part 3 (5 November 2026), Anthropic's Claude Code (Claude Opus 5.5) was used.

What Claude Code produced. It analysed the repository and wrote the twelve-document documentation package and this submission. It implemented the Google SSO endpoint and client, the FCM push sender and receiver, WorkManager reminders, the isiZulu and Setswana string catalogue (first-draft translations), fixes for the audit findings, the release-signing setup and the Play Store kit. It also added backend and Android unit tests, and drove an emulator to capture the screenshots and check offline sync and notifications.

How the output was checked. All code was compiled, and the tests were run: 29 backend, 42 Android unit and 16 instrumented, all passing (section 8). Behaviour was checked on an emulator. One defect found this way, a crash on opening Settings, was fixed before submission. Statements in the documents were checked against the code and test output; unknowns are labelled rather than guessed. Translations are marked for first-language review. **Team action** each member should review the changes, run the app on a phone, and edit this section to describe their own review.

Citation. Anthropic (2026) *Claude Code* [AI coding assistant], Claude Opus 5.5. Used 5 November 2026 for implementation, testing support, translation drafts and documentation. OpenAI (2026) *Codex*, used September 2026. Cline (2026) *Cline* VS Code extension, used September 2026. GitHub (2026) *Copilot coding agent*, used 23 September 2026. Prompts and outputs are kept in the tool session logs.

7 Publication preparation

Play Store readiness

Item	Evidence
Upload key and signing	`scripts/create-upload-keystore.ps1` created a 4096-bit RSA PKCS12 upload key outside the repository; `keystore.properties` is git-ignored. Release signing verification: On 2026-11-05 a release APK (R8 minify and resource shrinking) was built at 13:51 and checked with `apksigner verify --verbose --print-certs`. Result: "Verified using v2 scheme (APK Signature Scheme v2): true", 1 signer, certificate DN "CN=Nexus, OU=OPSC6312, O=Nexus, C=ZA", SHA-256 b100fcba2328e91298e79113379e16a45c9cf9b4af5f708f607fc16039508f9f (matches the upload key). That Gradle run then failed on a configuration-cache problem in the release HTTPS guard. After that was fixed, `assembleRelease` completed with BUILD SUCCESSFUL, and an http:// URL is still rejected. The verification builds used the placeholder https://api.nexus.invalid/api/ and were deleted: rebuild the submittable APK/AAB with the real HTTPS API URL.
Release build guard	Release builds fail unless `NEXUS_API_URL` is HTTPS; R8 minify and resource shrinking are on.
Store listing kit	`docs/play-store/` : 512 px icon (32-bit), 1024×500 feature graphic, 8 phone screenshots (1080×2160), listing text, Data safety answers, content rating, privacy policy.
Remaining	**Team action** host the API, publish the privacy policy, build `bundleRelease` with the real URL, upload to Play Console Internal testing, and screenshot the Console.

8 Test results (2026-11-05)

Verified locally; CI could not run (account billing lock)

Suite	Result
Backend: `npm test` (Jest + Supertest, Zod 4.6.7)	29/29 passed: `server.test.js` 11, `sync.test.js` 5, `part3.test.js` 13 (SSO, push, device tokens, data-file safety, email case)
Android unit: `gradlew test`	42/42 passed, including new `AuthErrorMessagesTest` (5), `ReminderPlannerTest` (2), `CsvRowTest` (2) and a `HabitScoringTest` case
Android instrumented: `connectedDebugAndroidTest`, Android 7.0 emulator	16/16 passed with the API stopped. Earlier runs: 14/16 with a live API (AUD-22) and one flaky instrumentation crash.
On-device walkthrough	Sign-in error, all screens, three languages, offline create → sync → present on the server, reminder ("1 overdue and 2 due today"), test notification, Settings crash fixed and re-verified

9 What the team must still do

1. Review all changes, commit, push, merge to `main`, close Dependabot PR #27 as superseded, and tag `final-poe`.
2. Resolve the GitHub Actions billing lock and confirm green CI on the final commit.
3. Host the API over HTTPS (any Node 22 or container host with a persistent disk; see NX-11-DEP), and set `JWT_SECRET`, `GOOGLE_CLIENT_ID`, `FIREBASE_SERVICE_ACCOUNT_BASE64` and `TRUST_PROXY`.
4. Create the Google Cloud OAuth clients (Web + Android with the SHA-1s) and the Firebase project (`google-services.json`).
5. Build the signed release with the real URL and install it on a physical phone.
6. Record the video (section 5), upload it unlisted, and add the link to the README and the cover.
7. Publish the privacy policy (fill in the contact email), upload to Play Console Internal testing, and screenshot it.
8. Have isiZulu and Setswana speakers review `docs/i18n/catalog.py`, then run `python docs/i18n/generate_strings.py`.
9. Confirm the User Defined 4/5 feature names, the student details, and the AI-use write-up.

10 Appendix: documentation package (v2.0)

The twelve deliverables follow. They were updated for v2.0, and each document's control table states its evidence baseline.

NEXUS · COMBINED SUBMISSION

Nexus Documentation Package

Twelve numbered deliverables for the Nexus Android app and Node.js API, compiled from repository evidence.

Package control

Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending

Package contents

ID	Deliverable	Scope
NX-01-PRD	Product Requirements	Purpose, users, scope, functional requirements, constraints and acceptance criteria for Nexus as implemented.
NX-02-TRD	Technical Requirements	Architecture, technology stack, interfaces, quality attributes and operational requirements.
NX-03-FLOW	Application Flow	Navigation map, key user journeys, system interactions and error/alternate flows.
NX-04-DES	Design Brief	Design goals, navigation, visual conventions, accessibility and responsive behaviour as evidenced in the app.
NX-05-DATA	Backend and Data Schema	Entities, relationships, key fields, storage locations and data flows on the server and the device.
NX-06-PLAN	Implementation Plan	Completed work with evidence, and remaining work ordered by dependency with priorities, estimates, acceptance criteria and tests.
NX-07-CICD	CI/CD and Release	Verified pipeline behaviour, build and test gates, run history, environments and the release checklist.
NX-08-AUD	Engineering Audit	Evidence-based findings with severity, confidence, affected area and recommended action, plus an addendum verifying earlier remediation.
NX-09-TDH	Testing, Deployment and Handover	Test strategy and verified results, deployment summary, operational handover, known limitations and runbooks.
NX-10-UM	User Manual	Task-based instructions for Nexus users: account, projects, tasks, habits, settings and working offline.
NX-11-DEP	Deployment Guide	How the API and app are actually deployed today (local), the container path the repository supports, and labelled recommendations for public hosting.
NX-12-TCR	Team Contribution Report	Verifiable contribution evidence from the repository and GitHub, with unavailable evidence marked as unknown.

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 01

Product Requirements

Purpose, users, scope, functional requirements, constraints and acceptance criteria for Nexus as implemented.

Document control

Document ID	NX-01-PRD
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Module assessor reviewing the PoE; developers and testers taking over the codebase
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-01-PRD contents

1. 1 Purpose and product summary
2. 2 Users and roles
3. 3 Scope
4. 4 Functional requirements
5. 5 Non-functional requirements
6. 6 Constraints and assumptions
7. 7 Success measures
8. 8 Open questions

1 Purpose and product summary

Version 2.0 update (Part 3). Requirements FR-21 to FR-23 (Google SSO, real-time push, due-date reminders) are new. FR-05 and FR-18 are now fully implemented, and the defects behind FR-01 and FR-02 are fixed. Each row states its current status.

Nexus is an Android app for personal project planning and habit tracking, backed by a small Node.js/Express REST API in `backend/`. The README describes it as "a calm, focused Android workspace for project planning and habit tracking". This document records what the product does at the evidence baseline, what it deliberately does not do, and how each requirement can be accepted. Requirement IDs (`FR-nn`, `NFR-nn`, `CON-nn`) are referenced by every other deliverable.

> **How to read status.** Implemented means the behaviour exists in source at the baseline commit; it does not by itself mean the behaviour was exercised end-to-end on a device for this package. Verified test results are listed in NX-09-TDH. Gaps are tracked as findings in NX-08-AUD and as tasks in NX-06-PLAN.

2 Users and roles

User roles evidenced in code

ID	Role	Description	Evidence
ROLE-01	Visitor (not signed in)	Can view the splash screen, register and sign in. Every other route redirects to Login.	`NexusNavHost.kt` auth guard Implemented
ROLE-02	Signed-in user	Owns their projects, tasks, habits and profile. All API data is partitioned by the JWT `sub` (user id); the Room cache is partitioned by the same owner id.	`server.js` `withAuthAndDb`; `Database.kt` `userId` columns Implemented

There is no administrator, collaborator or shared-project role. The README's test-credential table labels one seeded account "Global system administrator", but the API has no role field or role checks, so that account is an ordinary user Implemented. See AUD-09.

3 Scope

3.1 In scope (implemented)

- Account lifecycle: register, sign in (email/password or Google SSO), persisted session, sign out, profile, preferences, password change, account deletion.
- Notifications: scheduled due-date reminders and real-time push of changes made on another device.
- Projects with tasks (due date, priority, status, labels, checklist), search and filters.
- Habits with daily completion, streaks, points, levels, badges and CSV export.
- Offline-first local saves with an ordered upload queue, sync status banner and conflict review.
- English, isiZulu and Setswana on every screen; light/dark/system theme; animated splash that respects reduced motion.

3.2 Out of scope or planned (not implemented)

From `PROJECT_IMPROVEMENTS.md` ("Still planned") and `docs/POE_PART_2_ADDENDUM.md`: board/calendar/timeline views, recurring tasks, attachments/templates/archive, collaboration/roles/comments, account recovery, general

import/export, widgets, tablet layouts, and SQLite on the server. None of these exist in code `Implemented`. Google SSO, push notifications, reminders and full localisation were planned for the final PoE and are now implemented (FR-21 to FR-23, FR-18).

4 Functional requirements

Acceptance criteria are written so a tester can run them against the debug build and a local API (setup in NX-11-DEP).

Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria	Evidence / status
FR-01	Register an account with display name, email and password.	Blank fields, a malformed email or a password under 6 characters are rejected on the device before sending. API returns 201 and a JWT; a duplicate email returns 409 `EMAIL_ALREADY_REGISTERED`, which the app explains ("An account with this email already exists"). Emails are case-insensitive.	`AuthViewModel.kt`, `registerSchema` `Implemented`; fixed in v2.0 (AUD-18, `AuthErrorMessagesTest`)
FR-02	Sign in and keep the session across app restarts.	Valid credentials return a JWT (24 h expiry) stored in EncryptedSharedPreferences; relaunching the app opens the Dashboard. Wrong credentials (401 `INVALID_CREDENTIALS`) show "Email or password is incorrect." Sign-in requires the API; there is no offline sign-in.	`TokenManager.kt`, `tokenFor()` `Implemented`; verified on device 2026-11-05 (AUD-18 fixed)
FR-03	Sign out.	A confirmation dialog appears; confirming clears the token and returns to Login. Unsent changes stay on the device and upload only when the same account signs in again.	`LogoutConfirmationDialog.kt`, `docs/ONLINE_OFFLINE.md` `Implemented`
FR-04	View and edit the profile display name.	Profile shows display name and email; Settings saves a new display name via `PATCH /users/me`. A cached profile is shown offline.	`ProfileScreen.kt`, `UserRepository.kt` `Implemented`
FR-05	Account preferences: language and notification preference.	Changing language (en/zu/tn) or the reminders switch sends a partial `PATCH /users/me`. If the language update fails it is marked pending and retried. The reminders switch controls the notifications in FR-23.	`SettingsViewModel.kt` `Implemented`; AUD-10 fixed
FR-06	Choose System, Light or Dark theme.	Selection applies immediately at the activity root and persists on the device only (not synced).	`SettingsPreferences.kt`, `MainActivity.kt` `Implemented`
FR-07	Change password.	Requires the current password and a new password of 6 or more characters. Wrong current password returns 401 `INCORRECT_CURRENT_PASSWORD`; success returns 204. Requires the API.	`PasswordDialog.kt`, `/users/me/password` `Implemented`
FR-08	Delete account and all data.	A confirmation dialog ("This permanently deletes your account and synced data.") precedes deletion. The API removes the user, projects, tasks and habits; the app purges that account's local rows and queue, then signs out.	`SettingsScreen.kt`, `DELETE /users/me`, `NexusApp.kt` purge callback `Implemented`

ID	Requirement	Acceptance criteria	Evidence / status
FR-09	Dashboard ("Today") overview.	Shows counts for projects, tasks and activity (completed tasks), plus "Your schedule" cards: Overdue, Due today and the next 3 Upcoming incomplete tasks. Shows "Showing saved data" when served from cache. Refresh also downloads every project's tasks for offline use.	`DashboardScreen.kt`, `DashboardRepository.kt` **Implemented**
FR-10	Create, view, edit, delete and search projects.	A project needs a non-empty name; description is optional. Search filters the list by name. Deleting a project also deletes its tasks locally and on the server.	`ProjectListScreen.kt`, `ProjectDetailScreen.kt`, `ProjectRepository.kt` **Implemented**
FR-11	Create, edit, complete and delete tasks within a project.	A task has title (required), description, due date (yyyy-MM-dd), priority (None/Low/Medium/High), status (To do/In progress/Done), labels and checklist steps. Marking Done sets `isCompleted`; the API keeps `status` and `isCompleted` consistent.	`TaskEditorDialog.kt`, `taskSchema` **Implemented**
FR-12	Find tasks quickly.	Task list offers text search over title and labels, status chips (All, Open, In progress, Done) and a High priority chip; shows progress "x of y tasks complete". A task detail screen shows priority, status, labels and checklist with an Edit action.	`TaskListScreen.kt`, `TaskDetailScreen.kt` **Implemented**
FR-13	Create, edit, delete and complete habits.	A habit has a name and frequency (Daily, Weekly, Custom). "Complete today" toggles today's date in `completedDates`; the button reads "Completed today" when done.	`HabitsScreen.kt`, `HabitCard.kt`, `HabitsViewModel.kt` **Implemented**
FR-14	Habit progress and rewards.	Habits screen shows level, points, current and best streak, completion rate, badges (3-day streak, 7-day streak, Reliable habit builder) and reward suggestions, all computed on the device.	`HabitScoring.kt`, `HabitScoringTest.kt` **Implemented**; AUD-17 fixed in v2.0
FR-15	Export habit history as CSV.	Settings, then "Export habits CSV" builds `habit,frequency,completed_date` rows; "Share CSV" opens the Android share sheet with `text/csv`.	`SettingsViewModel.exportCsv()`, `CsvRowTest` **Implemented**; AUD-11 fixed (RFC 4180 quoting)
FR-16	Work offline and sync later.	Project, task and habit creates, edits and deletes are saved to Room together with an upload record in one transaction, and they survive an app restart. Uploads run on refresh, when the network returns, when you tap Sync, and every 30 s while the process is alive. The banner shows the pending count and state.	`SyncManager.kt`, `NexusApp.kt`, `OfflinePersistenceTest.kt` **Implemented**
FR-17	Handle sync conflicts without losing data silently.	Retried operations are never duplicated (server keeps operation receipts). An edit to an item deleted elsewhere returns 409 and stops the queue; Review offers "Keep changes" or "Discard item edits". Existing-item edits are last-upload-wins.	`/api/sync`, `sync.test.js` **Implemented**
FR-18	Use the app in English, isiZulu or Setswana.	Selecting a language (login screen or Settings) switches every screen, dialog, error, sync message and notification. Each locale has 210 strings and 4 plurals, generated from one catalogue. Translations need first-language review.	`docs/i18n/catalog.py`, `values-zu`, `values-tn`, `LocalizedMessages.kt` **Implemented**; verified on device (screenshots 14–18)

ID	Requirement	Acceptance criteria	Evidence / status
FR-19	Animated splash and motion that respect accessibility settings.	With system animations disabled, the splash and screen transitions render without animation.	`SplashIntroScreen.kt`, `Motion.kt`, `SplashIntroScreenTest.kt` Implemented
FR-20	Show server notices.	The Dashboard shows a banner for maintenance mode or the first announcement from `GET /app/config`. The API always returns maintenance off and no announcements.	`RemoteConfigViewModel.kt`, `/api/app/config` Implemented
FR-21	Sign in with Google (single sign-on).	"Continue with Google" appears on Login and Register when the build has a Web client ID. Credential Manager returns a Google ID token; `POST /api/auth/google` verifies signature, audience, expiry and email verification with Google, creates the account on first use (201) or links an existing email (200), and returns a Nexus JWT. Invalid tokens give 401; an unconfigured server gives 503, which the app explains. A Google-only account cannot use password sign-in (401) or change password (409).	`GoogleSignIn.kt`, `server.js`, `part3.test.js` (3 tests, stubbed verifier), `AuthErrorMessagesTest` Implemented. End to end with real Google: Unknown until the OAuth clients are created
FR-22	Real-time push notifications.	After sign-in the app registers its FCM token (`POST /users/me/devices`, at most 10 per user; a token belongs to one account). When a synced change is applied, the API pushes "Project "X" added on another device" to the user's other devices, excluding the sender. The receiving app shows the notification and refreshes its offline cache. Dead tokens are pruned. "Send test notification" in Settings triggers a server push and shows a local one.	`Push.kt`, `push.js`, `part3.test.js` (5 tests) Implemented; local display verified on device; FCM delivery Unknown until a Firebase project is configured
FR-23	Due-date reminders.	With reminders on, a WorkManager job runs about every 12 hours (and immediately after sign-in or switching reminders on). It reads the offline cache and posts "Tasks need your attention: n overdue and m due today" with up to three task titles. It runs while the app is closed and without network. Android 13+ asks for the notification permission.	`Reminders.kt`, `ReminderPlanner` + `ReminderPlannerTest` Implemented; verified on device ("1 overdue and 2 due today")

5 Non-functional requirements

Quality requirements evidenced in the implementation

ID	Requirement	Evidence / status
NFR-01	Credentials and tokens are protected: bcrypt (cost 12) password hashes; JWT secret of at least 32 characters enforced at startup; token stored encrypted on the device; auth endpoints (including Google sign-in) rate-limited to 10 requests per 15 min per client IP (proxy-aware with `TRUST_PROXY`); Google ID tokens verified server-side.	`server.js`, `TokenManager.kt`, `server.test.js` Implemented
NFR-02	Release builds use HTTPS only; cleartext HTTP is allowed only in debug builds.	`network_security_config.xml` (main/debug), `preReleaseBuild` check Implemented

ID	Requirement	Evidence / status
NFR-03	Data isolation: each account sees only its own server data and cached rows.	Per-user maps on the server; `userId`-scoped DAO queries; sync ownership test Implemented
NFR-04	Durability: a local save is never lost because an upload failed; server writes are atomic (temp file + rename) and serialised by a mutex.	`SyncManager.save()`, `saveDatabase()` Implemented; corrupt-file risk AUD-01
NFR-05	Accessibility: icon buttons carry content descriptions; reduced motion is respected; fallback brand colours were chosen for contrast.	Code review only Implemented; no formal accessibility audit has been run Unknown
NFR-06	Compatibility: Android 7.0 (API 24) and later; target/compile SDK 37.	`app/build.gradle.kts` Implemented
NFR-07	Responsiveness on poor networks: 5 s connect timeout and 15 s call timeout, after which cached data is shown.	`RetrofitClient.kt` Implemented
NFR-08	Observability: structured JSON logs (Pino) with request ids and redaction of passwords and authorization headers; `/health` liveness endpoint.	`server.js` Implemented

6 Constraints and assumptions

Constraints

ID	Constraint	Source
CON-01	The API stores all data in one JSON file and must run as a single instance; replicas must not share the file.	`docs/ONLINE_OFFLINE.md`, `server.js` Implemented
CON-02	No public API host has been provisioned. Release builds cannot be produced without an HTTPS `NEXUS_API_URL`.	README, `app/build.gradle.kts` Implemented
CON-03	Background upload is not guaranteed while the app process is stopped; there is no WorkManager job.	`docs/ONLINE_OFFLINE.md`, `NexusApp.kt` Implemented
CON-04	The project has no PostgreSQL or other database server, and the PoE material must not present one.	`docs/POE_PART_2_ADDENDUM.md` Implemented
CON-05	Data that was never downloaded to a device cannot appear offline.	`docs/ONLINE_OFFLINE.md` Implemented

7 Success measures

The repository defines no product metrics or analytics. `PROJECT_IMPROVEMENTS.md` asks that each feature define "one measurable success outcome", but none is recorded Unknown. Recommendation Before the final PoE, agree on two or three measures that can be checked during testing, for example "a task created offline appears on the server within one sync cycle of reconnecting" (FR-16).

8 Open questions

Decisions needed from the team

ID	Question	Related
OQ-01	*Resolved 2026-11-05:* reminders were built for the final PoE (FR-23).	FR-05, AUD-10, IMP-08
OQ-02	Which public host will run the API, and who owns its secret and backups? Open question	CON-02, NX-11-DEP
OQ-03	Is a "Global system administrator" role actually required? If so it needs server-side authorisation; if not, the README label should be corrected. Open question	ROLE-02, AUD-09
OQ-04	*Resolved:* all three are implemented (FR-21, FR-22, FR-18). Still open: who creates and owns the Google Cloud and Firebase projects. Open question	NX-06-PLAN

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 02

Technical Requirements

Architecture, technology stack, interfaces, quality attributes and operational requirements.

Document control

Document ID	NX-02-TRD
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Developers and operators maintaining Nexus; assessor checking technical depth
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-02-TRD contents

1. 1 System context
2. 2 Technology stack
3. 3 Architecture requirements
4. 4 Interfaces
5. 5 Quality attributes
6. 6 Operational requirements

1 System context

Version 2.0 update. Part 3 added Google SSO (Credential Manager plus `google-auth-library`), FCM push (`firebase-messaging` plus `firebase-admin`), WorkManager reminders, full en/zu/tn resources, release signing, Zod 4 and Node 22. The diagram shows the core components; the new pieces are listed in the tables below.

Nexus has two deployable parts: the Android app (`app/`, package `com.example.nexus`) and the Nexus API (`backend/server.js`). The app talks to the Nexus API over REST/JSON. Since v2.0 it also uses Google Identity (Credential Manager) for SSO and Firebase Cloud Messaging for push. A Retrofit client for the public MyMemory translation API is defined, but nothing in the app calls it (Implemented, AUD-09).

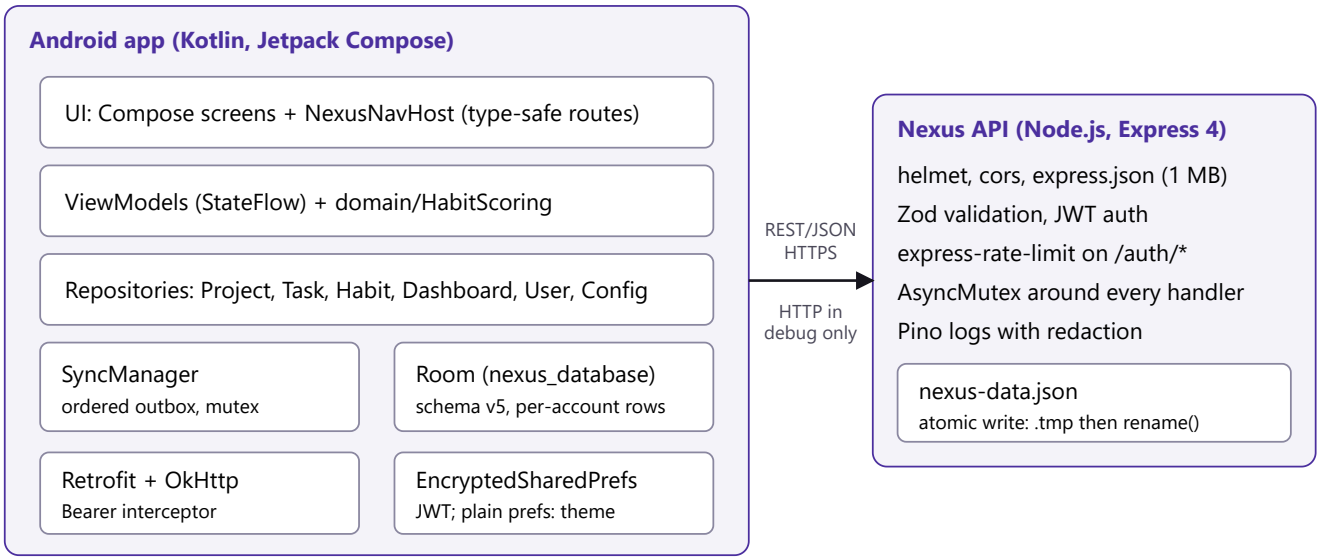


Figure 1. Component view at the evidence baseline. Arrows show request direction only.

2 Technology stack

Versions as declared in build files

Layer	Technology and version	Source
Android language / build	Kotlin 2.0.21, AGP 9.3.1, Gradle 9.7.1 wrapper, JDK 17 target, KSP 2.0.21-1.0.28	`build.gradle.kts`, `gradle-wrapper.properties`
SDK levels	minSdk 24, targetSdk 37, compileSdk 37; versionCode 1, versionName 1.0	`app/build.gradle.kts`
UI	Jetpack Compose (BOM 2026.02.01), Material 3, Navigation Compose 2.8.0 with kotlinx-serialization routes, material-icons-extended	`libs.versions.toml`, `app/build.gradle.kts`
Networking	Retrofit 3.0.0 + Gson converter, OkHttp logging-interceptor 4.12.0 (BASIC in debug, NONE in release)	`RetrofitClient.kt`

Layer	Technology and version	Source
Local storage	Room 2.8.5 (runtime, ktx, compiler via KSP); AndroidX Security Crypto 1.1.0	`Database.kt`, `TokenManager.kt`
SSO and notifications (v2.0)	androidx.credentials 1.6.0 + play-services-auth, googleid 1.1.1; WorkManager 2.12.0; Firebase BoM 35.1.0 (firebase-messaging); google-services plugin 4.5.0, applied only when `app/google-services.json` exists	`app/build.gradle.kts`
Quality tooling	ktlint plugin 12.1.2, detekt 1.23.8, Kover 0.9.1; JUnit 4.13.2, MockK 1.13.12, coroutines-test 1.8.1, Compose UI test	`build.gradle.kts`
API runtime	Node.js 22 (CI and Docker; required by google-auth-library 11 and firebase-admin 14), ES modules	`.github/workflows/*.yml`, `backend/Dockerfile`
API libraries	express ^4.21.2, bcryptjs ^2.4.3, jsonwebtoken ^9.0.2, zod ^4 (4.6.7 installed), google-auth-library ^11.3.3, firebase-admin ^14.5.0, express-rate-limit ^7.5.0, helmet ^8.3.0, cors ^2.8.6, pino ^10.3.1, pino-http ^11.0.0	`backend/package.json`
API tests	Jest ^29.7.0 and Supertest ^7.0.0, run in band with `--experimental-vm-modules`	`backend/package.json`

3 Architecture requirements

Technical requirements

ID	Requirement	How it is met
TR-01	Single-activity MVVM with feature packages.	`MainActivity` hosts `NexusNavHost`. UI is grouped under `ui/<feature>`; ViewModels expose `StateFlow` UI state **Implemented**.
TR-02	Repositories own all data access; ViewModels never call Retrofit directly.	Six repositories are created once in `NexusApp` (manual DI container). ViewModels take constructor parameters with defaults so tests can inject fakes **Implemented**.
TR-03	Business rules live outside the UI.	`domain/HabitScoring` is pure Kotlin and is unit-tested **Implemented**.
TR-04	Local writes are transactional with their outbox record.	`SyncManager.save()` runs the entity write and the `PendingChange` insert inside `withTransaction`, under a coroutine `Mutex` **Implemented**.
TR-05	Reads never overwrite unsent local edits.	`SyncManager.read()` flushes the queue first and falls back to Room if the flush fails or the network is unavailable **Implemented**.
TR-06	Room schema changes ship with migrations; no destructive fallback.	Migrations 3→4 and 4→5 are registered; `fallbackToDestructiveMigration` is deliberately absent **Implemented**. Schema export is off (AUD-12).
TR-07	API base URL is configured at build time.	`NEXUS_API_URL` Gradle property or environment variable must match `https?://host[:port]/path/`. It defaults to `http://10.0.2.2:5263/api/`, and release builds fail unless it starts with `https://` **Implemented**.
TR-08	API validates every write before it touches the store.	Zod schemas for register, login, profile (strict), password, project, task, habit and sync. Failures return 400 `VALIDATION_ERROR` **Implemented**.

ID	Requirement	How it is met
TR-09	API errors use one response shape.	Central handler returns `{ error: { message, requestId, code? } }`; in production a 500 message is replaced with "Internal Server Error" **Implemented**.
TR-10	Idempotent offline replay.	`POST /api/sync` takes `operationId` (UUID) and stores receipts per user. A repeated operation returns 204 without effect **Implemented**.

4 Interfaces

4.1 Nexus API endpoints

All paths are under `/api` except the root `/health`. "Auth" means a `Authorization: Bearer <JWT>` header is required; a missing or invalid token gives 401 `UNAUTHORIZED`, and a token for a deleted user gives 401 `SESSION_EXPIRED`.

Routes defined in backend/server.js

Method and path	Auth	Purpose and notable responses
`GET /health`, `GET /api/health`	No	`{status:"ok", uptime}`
`GET /app/config`	No	Static config: versions 1.0.0, maintenance off, feature flags, no announcements
`GET /localization/:language`	No	Five-string bundle for en/zu/tn (falls back to en)
`POST /auth/register`	No, rate-limited	201 `{token}`; 409 duplicate email; 429 after 10 per 15 min
`POST /auth/login`	No, rate-limited	200 `{token}`; 401 invalid credentials (emails case-insensitive)
`POST /auth/google`	No, rate-limited	`{idToken}`; 201 new account or 200 existing/linked, `{token}`; 401 `INVALID_GOOGLE_TOKEN`; 503 `SSO_NOT_CONFIGURED`
`POST /users/me/devices`, `DELETE /users/me/devices/:token`	Yes	Register or unregister an FCM token (204)
`POST /users/me/notifications/test`	Yes	Pushes a test notification; returns `{configured, sent, devices}`
`GET` / `PATCH /users/me`	Yes	Profile `{email, displayName, language, notificationsEnabled}`; PATCH rejects unknown fields
`POST /users/me/password`	Yes	204; 401 wrong current password
`DELETE /users/me`	Yes	204; removes the user and all their data
`GET /dashboard`	Yes	`{projects, tasks, activity}` counts
`POST /sync`	Yes	204 applied or duplicate; 400 invalid payload; 409 `SYNC_CONFLICT`
`GET` / `POST /projects`	Yes	List / create (201)
`GET` / `PUT` / `DELETE /projects/:projectId`	Yes	Read / update / delete (delete cascades to tasks); 404 `PROJECT_NOT_FOUND`

Method and path	Auth	Purpose and notable responses
`GET` / `POST /projects/:projectId/tasks`	Yes	List / create task; 404 if the project is not owned
`PUT` / `DELETE /tasks/:taskId`	Yes	Update / delete; 404 `TASK_NOT_FOUND`
`GET` / `POST /habits`, `PATCH` / `DELETE /habits/:habitId`	Yes	List / create / update / delete; 404 `HABIT_NOT_FOUND`

When `NODE_ENV=test` only, `GET /api/test/error-500` also exists for error-handler tests **Implemented**. The app uses the direct project, task and habit endpoints only when no `SyncManager` is wired in. In the production wiring (`NexusApp`), writes go through `POST /sync` **Implemented**.

4.2 Device integrations

- Permissions: `INTERNET`, `ACCESS_NETWORK_STATE` and (v2.0) `POST_NOTIFICATIONS`, requested at runtime on Android 13+ **Implemented**.
- WorkManager periodic job (reminders) and an FCM `FirebaseMessagingService` (v2.0) **Implemented**.
- `ConnectivityManager` network callback triggers sync when a network becomes available **Implemented**.
- Android share sheet for CSV export (`Intent.ACTION_SEND`, `text/csv`) **Implemented**.
- Backup: `allowBackup="true"`; since v2.0 both shared preferences and the Room database are excluded from cloud backup and device transfer (AUD-05 fixed).

5 Quality attributes

Quality attributes and current position

Attribute	Current position	Related
Security	Controls listed in NFR-01 and NFR-02 are implemented. CORS is open (`*`), which is documented as intentional for a mobile client. No formal security review or penetration test has been performed **Unknown**.	AUD-06, AUD-16
Reliability	Atomic file writes and an in-process mutex. However, a JSON file that cannot be parsed at startup is replaced on the next write (AUD-01).	NFR-04
Scalability	Single process and single file; every request, including bcrypt hashing, holds one global lock. `PROJECT_IMPROVEMENTS.md` treats this as adequate for fewer than 50 users and 10,000 tasks **Inferred**; this was not load-tested.	CON-01, AUD-07
Maintainability	Feature packages, shared test fakes, ktlint/detekt/Kover configured. Lint is not enforced in CI (AUD-04).	NX-07-CICD
Portability	Android 7.0+. The API runs on any Node host or as a container.	NX-11-DEP

6 Operational requirements

API runtime configuration (names only; values are never stored in the repository)

Variable	Required	Behaviour
`JWT_SECRET`	Yes	At least 32 characters and not a known placeholder, or the process exits with code 1. Keep it stable across restarts, or existing sessions become invalid.

Variable	Required	Behaviour
`PORT`	No (5263)	HTTP listen port.
`DATABASE_PATH`	No	Data file path; defaults to `backend/nexus-data.json`, and to `/data/nexus-data.json` in the Docker image.
`NODE_ENV`	No	`production` hides 500 messages; `test` disables listening and enables the test route.
`LOG_LEVEL`	No (info)	Pino log level.
`GOOGLE_CLIENT_ID`	For SSO	Web OAuth client ID(s), comma-separated, accepted as the ID-token audience. Unset disables `/auth/google` (503).
`FIREBASE_SERVICE_ACCOUNT_BASE64`	For push	Base64 of the Firebase service-account JSON (or use `GOOGLE_APPLICATION_CREDENTIALS`). Unset makes push a logged no-op.
`TRUST_PROXY`	Behind a proxy	Express trust-proxy setting (e.g. `1`) so rate limiting uses the client IP.

- **OPS-01** TLS termination must be provided by the host or a reverse proxy; the API itself serves plain HTTP **Implemented**.
- **OPS-02** The data file (or `/data` volume) must be persistent and backed up. The repository contains no backup tooling **Implemented**.
- **OPS-03** Run exactly one API instance per data file (CON-01).

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 03

Application Flow

Navigation map, key user journeys, system interactions and error/alternate flows.

Document control

Document ID	NX-03-FLOW
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Testers, developers and the assessor following the demo
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-03-FLOW contents

1. 1 Navigation map
2. 2 Key user journeys
3. 3 System interactions
4. 4 Error and alternate flows

1 Navigation map

Routes are type-safe `@Serializable` objects in `NexusNavHost.kt`. The app starts at `Splash`. An auth guard sends any signed-out user on a protected route to `Login`, and sends a signed-in user on Login/Register to `Dashboard` Implemented. A bottom navigation bar (Today, Projects, Habits, Settings) appears on those four top-level routes only. A sync status banner appears on every authenticated screen except Splash.

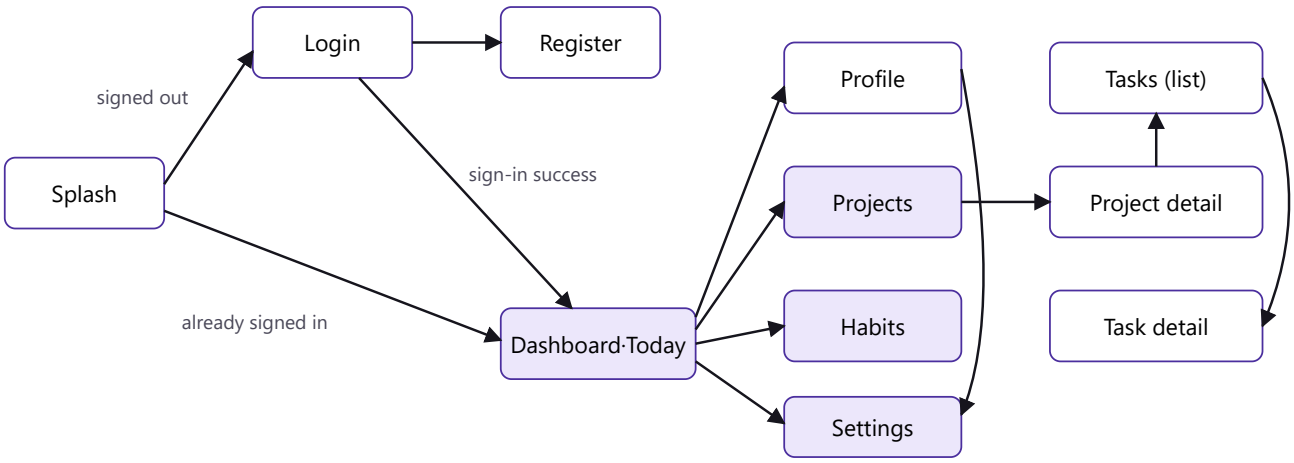


Figure 2. Navigation map. Shaded boxes are bottom-bar destinations. Back navigation pops the stack.

2 Key user journeys

Each journey lists the requirements it exercises. Steps reflect the implemented UI labels.

User journeys

ID	Journey	Main path	Requirements
UF-01	First launch and registration	Splash animation, then Login, then "Need an account? Register". Enter Display Name, Email and Password, tap Create Account. A JWT is saved and the Dashboard opens with an empty state, "Start your first project".	FR-01, FR-19
UF-02	Returning user	Splash, then Dashboard directly (token present). The Dashboard refresh uploads queued changes, then downloads projects and all their tasks into Room.	FR-02, FR-09, FR-16
UF-03	Plan a project	Projects tab, then the + (New Project) button. Enter a name, create, and open the project. Use "Add Task" or "Open Tasks Screen", then + (New Task): set title, due date, priority, status, labels and checklist, and save.	FR-10, FR-11
UF-04	Decide what to do today	The Today tab shows the Overdue / Due today / Upcoming cards. In a project's task list, use search, the status chips or "High priority"; open a task for its detail view; mark it complete.	FR-09, FR-12
UF-05	Daily habit check-in	Habits tab, then + (Add habit), name it and choose a frequency. Tap "Complete today" each day; stats (level, points, streaks, badges) recalculate on the device.	FR-13, FR-14

ID	Journey	Main path	Requirements
UF-06	Work offline	With the API unreachable, create or edit items as usual. The banner reads "Saved on device · waiting to sync" with "*n* pending changes". When the network returns (network callback, the 30 s loop, or tapping Sync), the queue uploads in order and the banner shows "All changes synced".	FR-16, TR-04
UF-07	Resolve a conflict	An upload is rejected with 409 (or 400). The banner shows "An item changed on another device · review required" with a Review button. The dialog offers "Keep changes" (the queue stays blocked and the edits are kept) or "Discard item edits" (that entity's queued operations and its local row are removed, then the rest of the queue retries).	FR-17, AUD-19
UF-08	Session expiry during sync	Upload returns 401, the banner says "Sign in again to sync saved changes", and a "Sign in" button clears the token and opens Login. Queued changes stay on the device for the same account.	FR-02, FR-16
UF-09	Manage account	Settings tab: edit Display Name; choose Appearance (System/Light/Dark) and Language; toggle reminders; "Sync account"; "Export habits CSV", then "Share CSV"; "Change password"; "Delete account and data" (confirm); "Log Out".	FR-04 to FR-08, FR-15
UF-11	Sign in with Google (v2.0)	Login or Register: tap "Continue with Google", pick an account in the Credential Manager sheet. The app posts the ID token to `/auth/google`, receives a Nexus JWT and opens Today. First use creates the account; an existing email account is linked.	FR-21
UF-12	Notifications (v2.0)	Settings: keep "Task due-date reminders" on (grant permission on Android 13+). A reminder lists overdue and due-today tasks. "Send test notification" shows a local notification and reports the server push result. Editing on phone A pushes "Task … updated on another device" to phone B, which refreshes its cache.	FR-22, FR-23
UF-10	Shared device, account switch	User A signs out with pending changes and user B signs in. B sees only B's rows. A's queue is neither shown nor uploaded until A signs in again.	FR-03, NFR-03

3 System interactions

3.1 Offline save and upload (SEQ-01)

1. The ViewModel calls a repository write, for example `TaskRepository.createTask()`. The repository generates a UUID for new entities on the device.
2. `SyncManager.save()` takes the mutex, then in one Room transaction writes the entity row and a `pending_changes` row (`operationId`, `resource`, `entityId`, `action`, JSON `payload`).
3. Later, `flushLocked()` reads the account's pending rows by `sequence` and posts each one to `POST /api/sync`. A row is deleted only after a 2xx response.
4. Server: if the `operationId` is already in the user's receipts it returns 204. Otherwise it validates the payload against the resource schema, checks for conflicts (update of a missing item, duplicate create, or a task whose project does not exist for this user), applies the change, records the receipt, saves atomically and returns 204. If the save fails, the in-memory change is rolled back.
5. The first failure stops the loop. Its HTTP status decides the banner message and whether Review is offered (400/409).

3.2 Read with cache fallback (SEQ-02)

1. `SyncManager.read()` takes the mutex and flushes pending changes first.
2. If the flush fails, Room data is returned, so unsent edits are never overwritten by server data.
3. If the flush succeeds, the remote call runs and replaces the account's cached rows in one transaction. On `IOException` the cache is returned and the banner reads "Offline · showing saved data". Other HTTP errors (for example 401) are passed up to the ViewModel.

3.3 Authentication (SEQ-03)

1. Login/Register posts credentials; the API checks them with bcrypt and returns a JWT with `sub` = user id, valid for 24 h.
2. `AuthSession.saveToken()` stores it encrypted. The cache owner id is decoded from the `sub` claim (not verified on the device; used only to scope the cache).
3. The OkHttp interceptor adds `Authorization: Bearer` to every Nexus API request.

4 Error and alternate flows

How failures surface to the user

ID	Condition	System behaviour	User sees
EF-01	API unreachable or timed out (5 s connect, 15 s call)	Writes are queued; reads fall back to Room.	Banner "Offline · changes saved on device" or "Offline · showing saved data"; Dashboard "Showing saved data". Sign-in shows "No internet connection. Check your network and try again."
EF-02	401/403 on a screen request	`ApiError.SessionExpired`; Settings clears the session.	"Your session has expired. Please log in again." Since v2.0, a wrong password at sign-in shows "Email or password is incorrect." instead (AUD-18 fixed).
EF-03	409 at registration (email exists) or 429 (rate limit)	Mapped by `AuthViewModel.authErrorMessage` (v2.0).	"An account with this email already exists. Sign in instead." / "Too many attempts. Please wait 15 minutes and try again."
EF-09	Google sign-in cancelled, no Google account on the device, or server SSO not configured	Cancel shows nothing; the other cases show a message.	"Add a Google account to this device…" / "Google sign-in is not available on this server yet."
EF-04	409/400 during sync	The queue stops at the failing operation; `conflict=true`.	Review button (UF-07)
EF-05	5xx from the API	`ApiError.ServerError`. In production the API hides internal messages.	"Something went wrong on our end. Please try again shortly." Sync banner: "Sync needs attention · saved changes kept on device".
EF-06	Item not available offline	Room has no row (never downloaded).	Error state with Retry, e.g. "Project is not available offline" (CON-05)
EF-07	Maintenance mode or announcement in `/app/config`	Dashboard banner. The API currently hard-codes maintenance off.	Maintenance message or the first announcement
EF-08	Delete account fails (offline or error)	The session is kept and nothing is purged.	Error message in Settings

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 04

Design Brief

Design goals, navigation, visual conventions, accessibility and responsive behaviour as evidenced in the app.

Document control

Document ID	NX-04-DES
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Designers and developers changing the UI; assessor reviewing UX decisions
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-04-DES contents

1. 1 Design goals
2. 2 Navigation model
3. 3 Visual conventions
4. 4 Accessibility
5. 5 Responsive behaviour
6. 6 Screen evidence

1 Design goals

The README's product principles are the stated design intent. The table shows how far the code follows each one.

Principles versus implementation

ID	Principle (README)	Where it shows up in the app
DS-01	Focus over clutter	The Today dashboard limits Upcoming to 3 tasks and shows "Nothing here." for empty cards **Implemented**.
DS-02	Fast capture	Floating + buttons on Projects, Tasks and Habits; dialogs need only a name or title; priority defaults to None and status to To do **Implemented**.
DS-03	Progress with context	Task list progress bar with "x of y tasks complete"; habit level, points and streaks **Implemented**.
DS-04	Calm by default	Short motion (150–300 ms), skeleton loaders instead of spinners, friendly empty states such as "A little progress, every day." **Implemented**
DS-05	Trustworthy data	A persistent sync banner always states whether changes are saved, pending, synced or in conflict **Implemented**.

2 Navigation model

- Bottom `NavigationBar` with four destinations: Today, Projects, Habits, Settings. Navigating between them saves and restores state and uses `launchSingleTop` **Implemented**.
- Second-level screens (Project detail, Tasks, Task detail, Profile) use a top app bar with a Back arrow ("Back" content description) **Implemented**.
- Create and edit actions use modal dialogs (project, task editor, habit, password), not separate screens **Implemented**.
- Destructive actions confirm first: log out, delete account and (since v2.0) deleting any project, task or habit. The project dialog warns that its tasks are deleted too **Implemented** (IMP-12).

The full map is in DIAG-02.

3 Visual conventions

3.1 Colour

Colour tokens from ui/theme/Color.kt and Theme.kt

Token	Value	Use
Brand purple / green	`#A855F7` / `#22C55E`	Logo gradient and wordmark
Light primary / secondary	`#6141B5` / `#236B54`	Fallback scheme on Android 11 and earlier (commit log: "accessible fallback brand colors")
Dark primary / secondary	`#D0BCFF` / `#22C55E`	Fallback dark scheme

Token	Value	Use
Background (light / dark)	`#F7F7FC` / `#12131B`	Overrides even when dynamic colour is on
Surface (light / dark)	`#FFFFFF` / `#191A24`	Cards and sheets

On Android 12 and later, Material You dynamic colour supplies primary, secondary and tertiary from the wallpaper, while background and surface stay fixed to the values above **Implemented**. Contrast was not measured for this package **Unknown**.

3.2 Shape, spacing and type

- Shapes: small 12 dp, medium 20 dp, large 28 dp rounded corners **Implemented**.
- Spacing scale in `Spacing.kt`: 2, 4, 6, 8, 10, 12, 16, 20, 22, 24, 32 dp. Top-bar logo 32 dp; auth logo 80 dp **Implemented**.
- Typography: the system sans-serif for every Material 3 style; body large 16 sp, body medium 14 sp **Implemented**. No custom font is bundled.
- Components: `NexusPrimaryButton`, `NexusSecondaryButton`, `NexusDestructiveIconButton`, `StatCard`, `Skeleton`, `FeedbackState` (empty/error with Retry), `NexusLogo` **Implemented**.

3.3 Motion

`Motion.kt` fixes durations: screen enter 220 ms, exit 180 ms, list item 220 ms, fade 150 ms, completion bounce 90 + 140 ms, success check 260 ms. Screen transitions slide a quarter width and fade **Implemented**.

4 Accessibility

Accessibility evidence

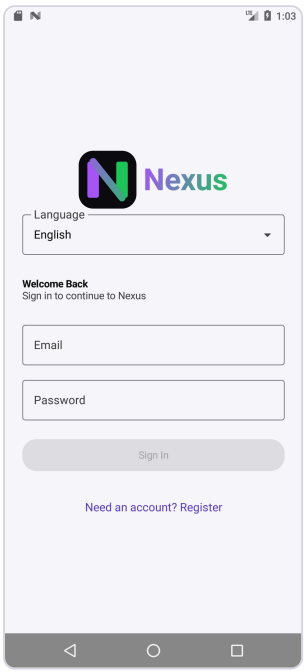
Aspect	Evidence	Status
Reduced motion	`rememberReduceMotion()` checks `ValueAnimator.areAnimatorsEnabled()`; navigation transitions, list reflow, shimmer and splash are skipped. `SplashIntroScreenTest` covers the splash.	**Implemented**
Screen reader labels	Icon buttons have descriptions such as "Back", "New Project", "Add habit", "Edit habit", "Remove label …", "Clear due date". Bottom-bar icons use `null` descriptions, with the visible text label as the accessible name.	**Implemented**
Not relying on colour alone	Priority and status are shown as text chips ("Priority · High"), not colour only.	**Implemented**
Keyboard / IME	`windowSoftInputMode="adjustResize"`; auth forms scroll above the keyboard (improvements log).	**Implemented**
Contrast, touch targets, TalkBack walkthrough	No measurements or audit records in the repository.	**Unknown** → IMP-11
Localised content	Every screen in en/zu/tn (210 strings each), including content descriptions, errors and notifications. Bottom-bar labels are single-line, so the longest isiZulu label ("Amaphrojekthi") fits.	**Implemented** (FR-18)

5 Responsive behaviour

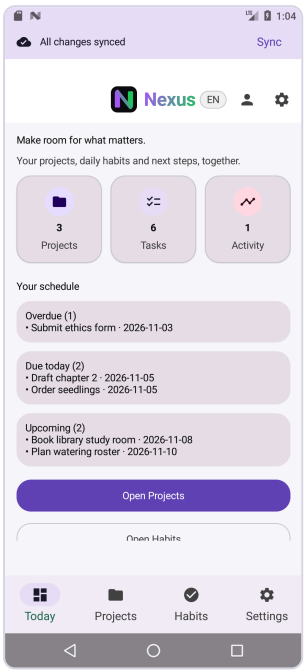
Layouts are single-column phone layouts built with `fillMaxWidth`, `LazyColumn` and horizontally scrolling chip rows. Window insets are handled with `statusBarsPadding`. No window size classes or tablet layouts exist; tablet layouts are listed as planned **Implemented**. Landscape behaviour has not been tested **Unknown**.

6 Screen evidence

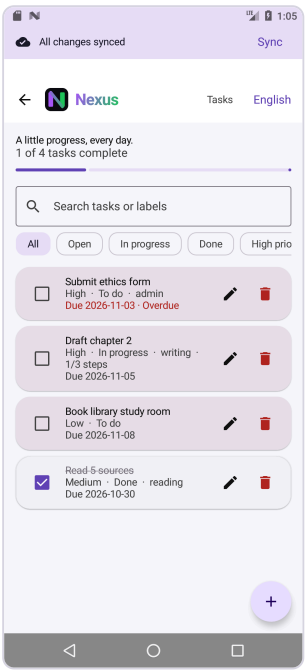
These screenshots were captured on 2026-11-05 from version 2.0.0 running on an Android 7.0 (API 24) emulator against the local API, using fictional demo data. They replace the old UI-audit set, whose file names did not match their content (AUD-21). All 21 are in `docs/screenshots/part3/`.



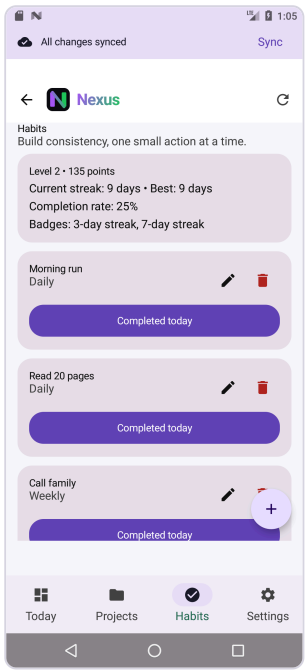
Login



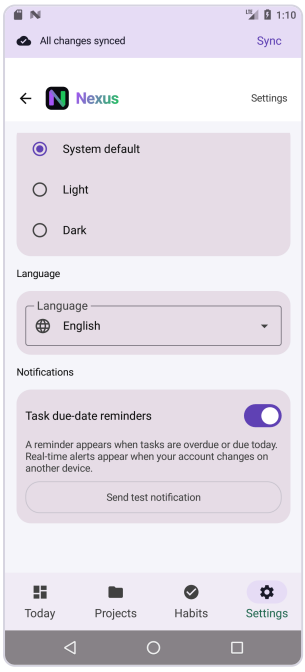
Today



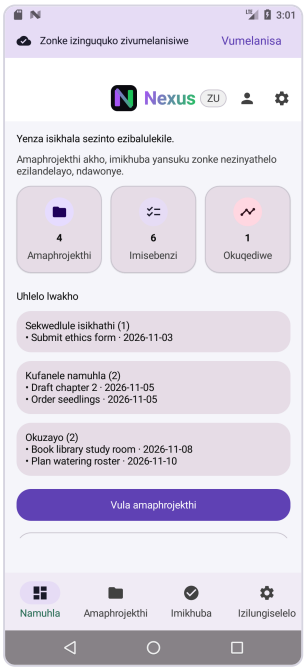
Tasks



Habits



Settings



isiZulu

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 05

Backend and Data Schema

Entities, relationships, key fields, storage locations and data flows on the server and the device.

Document control

Document ID	NX-05-DATA
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Developers changing the API or Room schema; assessor checking the data design
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-05-DATA contents

1. 1 Storage overview
2. 2 Server data model
3. 3 Device (Room) schema, version 5
4. 4 Data flows
5. 5 Data retention and privacy notes

1 Storage overview

Where Nexus data lives

Store	Location	Contents	Protection
API document store	`DATABASE_PATH` (default `backend/nexus-data.json`; `/data/nexus-data.json` in the container)	users, projects, tasks, habits, sync receipts	File-system permissions only; not encrypted at rest. Git-ignored.
Room database	App-private `nexus_database`, schema version 5	Cached projects, tasks, habits, dashboard counts, pending upload queue	App sandbox; excluded from Android backup since v2.0 (AUD-05 fixed)
Encrypted preferences	`nexus_secure_prefs`	JWT (`jwt_token`)	AES-256 GCM/SIV via AndroidX Security; excluded from backup
Plain preferences	Settings store; `cached_profiles`	`theme_mode`, `notifications_enabled`, `language`, language-sync flag; profile JSON per owner (`profile:<owner>`)	App sandbox; excluded from backup

2 Server data model

`server.js` keeps four in-memory `Map`s, persisted as one JSON document `{ users: [...], projects: {userId: [...]}, tasks: {userId: [...]}, habits: {userId: [...]} }`. Ownership is structural: every collection is keyed by the owner's user id, so there are no cross-user references Implemented.

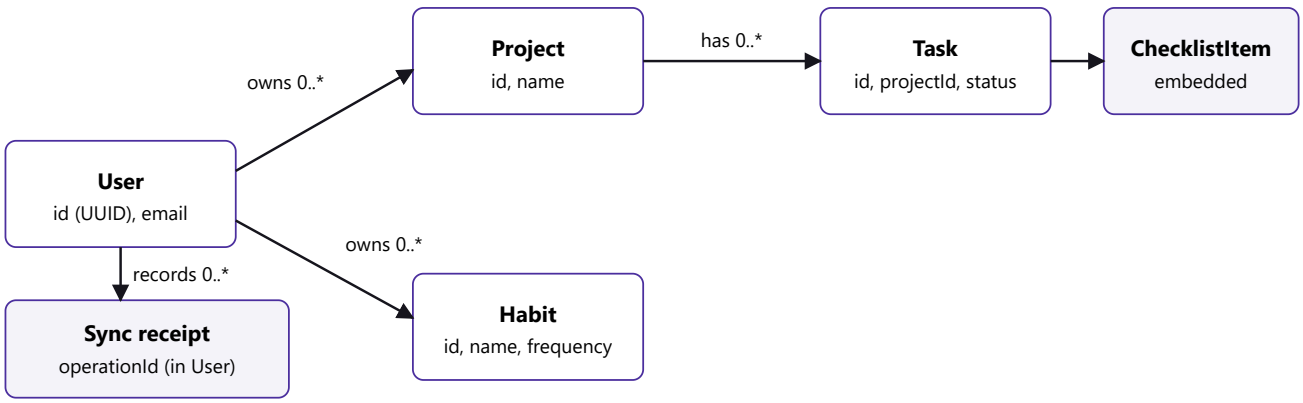


Figure 3. Logical entities. Tasks are stored per user and linked to a project by `projectId`; checklist items and labels are embedded arrays.

2.1 User (ENT-USER)

Field	Type	Rules
`id`	UUID string	`randomUUID()` at registration; JWT `sub`

Field	Type	Rules
`email`	string	Valid email, stored lower-case since v2.0; unique and matched case-insensitively **Implemented**
`displayName`	string	Non-empty
`passwordHash`	string	bcrypt cost 12. A non-bcrypt value is compared as plain text once and then rehashed (AUD-16). Never returned by the API.
`language`	string	Default `"en"`; any string accepted by the API, the app sends en/zu/tn
`notificationsEnabled`	boolean	Default `true`
`syncOperations`	string[]	Applied `operationId`s; grows without limit (AUD-07)
`googleSub` (v2.0)	string	Google account subject; set on first Google sign-in. Accounts created by Google have no `passwordHash`.
`deviceTokens` (v2.0)	string[]	FCM registration tokens, at most 10, newest last; a token is removed from any other account when registered

2.2 Project (ENT-PROJECT)

Field	Type	Rules
`id`	UUID string	Server-generated (REST) or device-generated (sync)
`name`	string	Non-empty
`description`	string | null	Optional
`createdAt`, `updatedAt`	ISO-8601 timestamp	Set by the server

Deleting a project deletes its tasks (REST and sync paths) **Implemented**.

2.3 Task (ENT-TASK)

Field	Type	Rules
`id`, `projectId`	UUID strings	The project must belong to the same user (404 on REST, 409 on sync)
`title`	string	Non-empty
`description`	string | null	Optional
`dueDate`	string | null	Pattern `yyyy-MM-dd` (format only; the calendar date is not checked)
`priority`	enum	`NONE` (default), `LOW`, `MEDIUM`, `HIGH`
`status`	enum	`TODO` (default), `IN_PROGRESS`, `DONE`
`isCompleted`	boolean	Derived: always `status === "DONE"` after a write
`labels`	string[]	Free text
`checklist`	{id?, title, isCompleted?}[]	Title non-empty

Field	Type	Rules
`createdAt`, `updatedAt`	ISO-8601	Server-set

2.4 Habit (ENT-HABIT)

Field	Type	Rules
`id`	UUID string	
`name`	string	Non-empty
`description`	string | null	Optional
`frequency`	enum	`DAILY` (default), `WEEKLY`, `CUSTOM`
`targetDays`	number[]	Accepted but not used by the app's scoring
`completedDates`	string[]	`yyyy-MM-dd` by convention; no format validation on the server
`createdAt` (+ `updatedAt` via sync)	ISO-8601	

3 Device (Room) schema, version 5

Room entities in data/Database.kt and data/SyncManager.kt

Table	Primary key	Columns
`projects`	`id`	`userId`, name, description, createdAt, updatedAt
`tasks`	`id`	`userId`, projectId, title, description, isCompleted, dueDate, priority, status, labels (JSON), checklist (JSON), createdAt, updatedAt
`habits`	`id`	`userId`, name, description, frequency, targetDays (JSON), completedDates (JSON), createdAt
`dashboard`	`userId`	projects, tasks, activity (counts)
`pending_changes`	`sequence` (autoincrement)	`userId`, operationId (UUID), resource (projects|tasks|habits), entityId, action (create|update|delete), payload (JSON)

List columns are stored as Gson JSON strings by `TaskConverters`. `userId` is the cache owner id: the JWT `sub`, or a hash of the token if no `sub` exists. No foreign keys or indices are declared **Implemented**.

3.1 Migration history

Migration	Change	Data effect
1→2, 2→3	Not present in code (version 2 added task fields, per commit 8b31aa3)	Upgrading from v1 or v2 would fail to open the database (AUD-12)
3→4	Adds `userId` to projects, tasks and habits; rebuilds `dashboard` keyed by user	Deletes all cached rows (they could not be attributed to an owner)
4→5	Creates `pending_changes`	Keeps existing rows

4 Data flows

Principal data flows

ID	Flow	Path
DF-01	Local write	ViewModel → Repository → `SyncManager.save()` → Room entity + `pending_changes` (one transaction). See SEQ-01.
DF-02	Upload	`pending_changes` (ordered) → `POST /api/sync` → in-memory map → `saveDatabase()` (`.tmp` then `rename`) → receipt stored → queue row deleted
DF-03	Refresh	`GET /projects` + `GET /projects/:id/tasks` for each project (N+1 requests) → replaces the account's cached projects and tasks → dashboard counts computed on the device
DF-04	Account deletion	`DELETE /users/me` → server removes the user and all collections → app purges that owner's Room rows and queue under the sync mutex
DF-06	Real-time push (v2.0)	Successful `/api/sync` → `notifyUser()` → FCM multicast to the user's other device tokens → `NexusMessagingService` shows a notification and refreshes the Room cache
DF-07	Google SSO (v2.0)	Credential Manager → Google ID token → `/api/auth/google` → Google key verification → user found/created/linked → Nexus JWT
DF-05	Export	Habits (cache or API) → CSV string in memory → Android share sheet. Nothing is written to shared storage.

5 Data retention and privacy notes

- Personal data held: email, display name, password hash, the user's own content and, since v2.0, the Google account subject and FCM device tokens. Google (sign-in) and Firebase (push delivery) act as service providers; no analytics or ads **Implemented**.
- Retention: data is kept until the user deletes it or the account. There are no server-side backups or expiry **Implemented**.
- Logs: Pino redacts password fields, `passwordHash` , `token` and the authorization header. pino-http's default serializers do not log request bodies, so emails in bodies do not reach the logs, but remote IP addresses are logged **Implemented**.

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 06

Implementation Plan

Completed work with evidence, and remaining work ordered by dependency with priorities, estimates, acceptance criteria and tests.

Document control

Document ID	NX-06-PLAN
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Development team planning the final PoE; assessor reviewing project management
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-06-PLAN contents

1. 1 Completed work
2. 2 Remaining work
3. 3 Definition of done
4. 4 Status at v2.0 (2026-11-05)

1 Completed work

This history is reconstructed from `PROJECT_IMPROVEMENTS.md`, merged pull requests and commit messages. Dates are commit or merge dates **Implemented**.

Delivered increments

Date	Increment	Evidence	Requirements
2026-08-23	Initial Android app with auth integration, secure token storage, nav flow, splash animation	Commits 65385b2, 0496717, 964cbf5	FR-01, FR-02, FR-19
2026-09-22	CI workflow, Room read cache, task due date / priority / status / labels / checklist, dashboard insights, auth validation	PRs #10–#16; commit 8b31aa3	FR-09, FR-11
2026-09-23	Backend remediation: mutex + atomic writes, Zod validation, JWT secret check, bcrypt, rate limiting, health check, Pino; task detail screen	PR #23; `server.test.js`	NFR-01, NFR-04, FR-12
2026-09-24	Android structural remediation: six repositories, manual DI, `HabitScoring` domain, feature packages, type-safe routes, shared fakes	PR #24	TR-01 to TR-03, FR-14
2026-09-26	UI refinement: theme colours, logo component	PR #25	NX-04-DES
2026-11-05	**Part 3 / v2.0:** Google SSO, FCM push, WorkManager reminders, full zu/tn localisation, audit fixes (AUD-01/02/03/05/06/10/11/16/17/18/21), Settings crash fix, release signing, Play Store kit, recaptured screenshots	`716bedb` + uncommitted working tree	FR-21, FR-22, FR-23, FR-18
2026-10-04	Offline upload queue and `/api/sync`, sync banner, conflict review, project search, task filters, zu/tn resources, configurable `NEXUS_API_URL`, HTTPS-only release, Dockerfile/compose	PR #26 (commit 3a3f275)	FR-16, FR-17, FR-18, TR-07

2 Remaining work

Tasks are ordered so that each depends only on tasks above it. Priority: **P0** blocks a credible release or submission, **P1** is needed before real users, **P2** is improvement. All effort figures are **Estimate** person-day estimates by the document author, not team commitments, and should be re-estimated by the team.

Dependency-ordered backlog

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-01	Restore GitHub Actions execution (resolve the account billing lock shown in run annotations) and re-run CI on `main`	P0	None	0.25 **Estimate**	Android CI and Backend CI both green on `main` at or after 3a3f275	Existing CI jobs (CI-01, CI-02)
IMP-02	Fix Zod 4 compatibility before merging PR #27: use `error.issues`; give `z.record` a key schema	P0	IMP-01	0.5 **Estimate**	Backend tests pass on Zod 4; validation still returns 400 `VALIDATION_ERROR`	`npm test`; existing Zod tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-03	Fail safe on an unreadable data file: refuse to start (or start read-only) instead of starting empty	P0	None	0.5 Estimate	A corrupt `nexus-data.json` stops the process with a clear log line and leaves the file untouched	New Jest test with a corrupt fixture (AUD-01)
IMP-04	Correct the auth error messages: 401 at login gives "Email or password is incorrect"; 409 gives "Email already registered"; 429 gives "Too many attempts"	P0	None	0.5 Estimate	Each status shows its own message; session expiry still shows the expiry text elsewhere	`AuthViewModelTest` cases per status (AUD-18)
IMP-05	Remove or fix the root `docker-compose.yml`; document `backend/compose.yaml` as the single container path	P1	None	0.25 Estimate	`docker compose up` from the documented path persists data across restarts	Backend CI smoke test plus a manual restart check (AUD-02)
IMP-06	Add `app.set("trust proxy", 1)` (configurable) for TLS-proxy hosting	P1	IMP-05	0.25 Estimate	The rate limit keys on the client IP behind the chosen proxy	Supertest with `X-Forwarded-For` (AUD-06)
IMP-07	Provision the hosted API and release signing; set the `NEXUS_API_URL` repository variable	P1	IMP-01, IMP-03, IMP-05, IMP-06	1–2 Estimate	`/health` reachable over HTTPS; signed release APK/AAB produced by CI	Release checklist in NX-07-CICD (AUD-08)
IMP-08	Due-date reminders, or hide the switch (decision OQ-01)	P1	OQ-01	0.25 (hide) / 3–5 (build) Estimate	The switch either has a visible effect or is not shown	Unit test on the scheduler, or UI test that the switch is absent (AUD-10)
IMP-09	Exclude the Room database from backup, or document the choice	P1	None	0.25 Estimate	`backup_rules.xml` and `data_extraction_rules.xml` exclude `nexus_database`	Manual `adb shell bmgr` check (AUD-05)
IMP-10	Enforce lint in CI (`ktlintCheck`, `detekt`) and fix the ktlint parse failure; delete the placeholder `blank.yml` and `build-verification.log`	P1	IMP-01	1 Estimate	CI fails on lint violations; the log file is no longer tracked	CI run (AUD-13)
IMP-11	Accessibility pass: TalkBack walkthrough, contrast measurement, 48 dp targets	P1	None	1–2 Estimate	Results recorded; issues fixed or logged	Accessibility Scanner report; Compose semantics tests

ID	Task	Pri.	Depends on	Est. (days)	Acceptance criteria	Tests
IMP-12	Undo for list deletes of projects, tasks and habits	P2	None	1 Estimate	Snackbar "Undo" restores the item before upload	ViewModel tests
IMP-13	CSV export escaping (RFC 4180 quoting)	P2	None	0.25 Estimate	A habit named `Read, write` exports as one quoted field	`SettingsViewModelTest` (AUD-11)
IMP-14	Fix the habit completion-rate formula: count habit-date pairs	P2	None	0.25 Estimate	Two daily habits both done for 30 days gives 100%	`HabitScoringTest` (AUD-17)
IMP-15	Enable Room `exportSchema` and add `MigrationTestHelper` tests for 3→4→5	P2	None	1 Estimate	Schema JSON committed; migration tests pass	Instrumented tests (AUD-12)
IMP-16	Documentation corrections: README contents links, admin role label, translation claim, recaptured screenshots	P2	IMP-07	0.5 Estimate	Every README link resolves; screenshots match their file names	Link check (AUD-09, AUD-21)
IMP-17	Complete feature-screen localisation (zu/tn)	P2	None	3–5 Estimate	No hard-coded user-facing English strings remain in `ui/`	Lint `HardcodedText`; pseudo-locale run (FR-18)
IMP-18	Final-PoE features (Google SSO, push notifications), as decided in OQ-04	P2	IMP-07, OQ-04	Unknown	Defined per feature in `docs/POE_PART_2_ADDENDUM.md`	Server and end-to-end tests required by the addendum

Total for P0 and P1 excluding IMP-08 "build" and IMP-18: about 5–7 person-days Estimate.

3 Definition of done

Adopted from the quality checklist in `PROJECT_IMPROVEMENTS.md`: API models and validation first; a Room migration for any schema change; loading, offline, retry, unauthorised and validation states covered; ViewModel unit tests plus instrumented coverage of the main path; accessibility checked; README and the improvements log updated Implemented. Recommendation Add "CI green on the PR" once IMP-01 is done.

4 Status at v2.0 (2026-11-05)

Backlog status after the Part 3 work

Task	Status	Evidence or remaining action
IMP-01	Open (team)	Needs the GitHub account billing lock resolved; then re-run CI. Workflows moved to Node 22.
IMP-02	Done	Zod 4.6.7 adopted; 29/29 backend tests pass. PR #27 can be closed as superseded.
IMP-03	Done	Refuses to start; `.bak` copy kept; 3 tests in `part3.test.js`.

Task	Status	Evidence or remaining action
IMP-04	Done	`AuthErrorMessagesTest` ; verified on device.
IMP-05	Done	Root `docker-compose.yml` uses a named volume at `/data` ; `docker compose config` valid. Not run (no Docker daemon).
IMP-06	Done	`TRUST_PROXY` env; not covered by a test.
IMP-07	Partly done	Signing config, upload key and Play kit done. API hosting and Play Console upload are team actions.
IMP-08	Done (built)	Reminders and push implemented (FR-22, FR-23).
IMP-09	Done	Backup rules exclude `database` .
IMP-10	Done (ktlint)	Codebase brought to zero ktlint violations (592 at the start: auto-format, line wrapping, constant renames, and two rules disabled in `.editorconfig` with recorded reasons). `ktlintCheck` now runs in Android CI before the tests. `blank.yml` and `build-verification.log` removed. detekt is not gated because the tool crashes in this toolchain (AUD-23).
IMP-11	Open	No TalkBack or contrast audit recorded.
IMP-12	Done (alternative)	Every delete (project, task, habit) now asks for confirmation, and the project dialog warns that its tasks are deleted too. Verified on device (screenshot 21). Undo was not implemented.
IMP-13	Done	`CsvRowTest` .
IMP-14	Done	New `HabitScoringTest` case.
IMP-15	Open	Room schema export and migration tests not added.
IMP-16	Done	README rewritten (contents links fixed, release notes, AI use); screenshots recaptured.
IMP-17	Done	210 strings × 3 languages; native-speaker review still recommended.
IMP-18	Done (code)	SSO and push implemented; need Google Cloud and Firebase projects to run end to end.

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 07

CI/CD and Release

Verified pipeline behaviour, build and test gates, run history, environments and the release checklist.

Document control

Document ID	NX-07-CICD
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Developers and whoever performs releases; assessor checking delivery automation
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-07-CICD contents

1. 1 Pipeline inventory
2. 2 Verified run history
3. 3 Local verification for this package
4. 4 Environments
5. 5 Release process

1 Pipeline inventory

Three GitHub Actions workflows live in `.github/workflows/`. Every workflow sets `permissions: contents: read` Implemented. No deployment (CD) job exists; artifacts are built and uploaded, but nothing is deployed Implemented.

Workflows

ID	Workflow	Triggers	Jobs and gates
CI-01	`android-ci.yml` "Android CI"	Push to `main`, `master`, `release/**`; PR to main/master; manual	**backend-test** (15 min): Node 22 (since v2.0; was 20), `npm ci`, `npm test` with a test-only JWT secret. **build** (30 min): JDK 17 Temurin, Gradle setup, wrapper validation, `ktlintCheck` (since v2.0), `./gradlew test`, `:app:assembleDebug`, upload `nexus-debug-apk` (14 days). Only if repository variable `NEXUS_API_URL` is set: `assembleRelease bundleRelease` and upload `nexus-release-artifacts`.
CI-02	`backend-ci.yml` "Backend CI"	Same branches, filtered to `backend/**` and its own file; manual	Node 22 tests, then `docker build`, run the container with `NODE_ENV=production`, wait 3 s, and require HTTP 200 from `/health`; the container is always removed.
CI-03	`blank.yml` "CI"	(removed in v2.0)	GitHub starter template that only echoed text; deleted (AUD-13).

Not gated in CI: detekt (AUD-23), Kover coverage thresholds, instrumented (`androidTest`) tests, and dependency vulnerability scanning Implemented. Dependabot opened PR #27, but the repository has no `.github/dependabot.yml`, so its configuration lives in GitHub settings Unknown.

2 Verified run history

Retrieved with `gh run list` / `gh run view` on 2026-11-05 (78 most recent runs: 49 success, 29 failure).

Notable runs

Date (UTC)	Trigger	Result	Cause (from run logs / annotations)
2026-09-22	Several pushes/PRs	Android CI mixed pass/fail	Early iterations; not analysed further
2026-09-24 07:17	Push to main (PR #24 merge)	**Android CI success**	Last successful Android CI on `main`
2026-09-26 18:23	PR #25 and its merge	All workflows failed in 3–5 s	Annotation: "The job was not started because your account is locked due to a billing issue."
2026-10-04 18:13	PR #26 and merge (baseline 3a3f275)	All workflows failed in 3–4 s	Same annotation; jobs never started. **The baseline code has never been verified by CI.**
2026-10-31 08:37	Dependabot PR #27 (zod 3.25.76 → 4.6.6)	Android build job passed; both backend test jobs **failed**	`TypeError: Cannot read properties of undefined (reading 'map')` at `server.js:133`. The Docker smoke test did not run. See AUD-03.

The Backend CI Docker smoke test (CI-02) has no successful run in the retrieved history Implemented. Its correctness is Inferred from reading the workflow only.

3 Local verification for this package

Commands run on 2026-11-05 on the author's Windows workstation

Command	Result
`cd backend; npm test` at v1.0 (zod 3.25.76)	2 suites, **16/16 passed**, 31.7 s
`cd backend; npm test` at v2.0 (zod 4.6.7)	3 suites, **29/29 passed**; `npm audit --omit=dev` : 0 vulnerabilities
`./gradlew :app:testDebugUnitTest --rerun` at v1.0 (JDK 21 runtime)	BUILD SUCCESSFUL; 7 classes, **32/32 passed**
`./gradlew :app:testDebugUnitTest` at v2.0	BUILD SUCCESSFUL; 10 classes, **42/42 passed**
`./gradlew :app:connectedDebugAndroidTest` at v2.0 (Android 7.0 emulator)	**16/16 passed** with the API stopped; 14/16 when a live API was reachable (AUD-22)
`assembleRelease -PNEXUS_API_URL=https://api.nexus.invalid/api/` at v2.0	APK signed with the upload key, verified by `apksigner` (APK Signature Scheme v2). A configuration-cache error in the HTTPS guard was fixed, after which the build is SUCCESSFUL; an http:// URL is rejected. The placeholder build was deleted.
`assembleDebug` at v2.0	Succeeded; installed and exercised on the emulator
Docker build	Not run Unknown. The Docker daemon was not running; `docker compose config` validates both compose files.

Details per suite are in NX-09-TDH.

4 Environments

Known environments

Environment	API URL	Status
Local emulator	`http://10.0.2.2:5263/api/` (debug default)	Implemented Documented in `docs/ONLINE_OFFLINE.md`
Local phone on LAN	`-PNEXUS_API_URL=http://<LAN-IP>:5263/api/`	Implemented Debug builds only
CI	None (tests use Supertest in process; smoke test uses localhost)	Implemented
Staging / production	None	Implemented Not provisioned; the README states the earlier production hostname "was not a configured service" (CON-02)

5 Release process

No release has been published: `gh release list` returns nothing and `versionCode` is 1 **Implemented**. The Gradle build enforces two release rules: `NEXUS_API_URL` must start with `https://` (`preReleaseBuild` check), and release builds minify and shrink resources **Implemented**. Since v2.0 a `release` signing config reads `keystore.properties` (git-ignored) or `NEXUS_KEYSTORE_*` environment variables, and the upload key was created with `scripts/create-upload-keystore.ps1` (AUD-08 fixed). CI does not have the key yet, so CI release artifacts remain unsigned until the secrets are added.

5.1 Release checklist

Items marked **Recommendation** are recommendations for the first release. The others restate rules the repository already enforces or documents.

1. CI (CI-01, CI-02) is green on the release commit; this requires IMP-01. **Recommendation**
2. Backend tests pass locally: `npm ci && npm test`.
3. Android unit tests pass: `./gradlew test`. Instrumented tests pass on at least one API 24 and one current device. **Recommendation**
4. Run the offline acceptance check in `docs/ONLINE_OFFLINE.md` (six steps) against the target API.
5. Increment `versionCode` / `versionName` in `app/build.gradle.kts` (2 / 2.0.0 for the final PoE). Settings shows `BuildConfig.VERSION_NAME`.
6. Hosted API is healthy over HTTPS with persistent storage and a stable `JWT_SECRET` (NX-11-DEP).
7. Set repository variable `NEXUS_API_URL` to the HTTPS URL ending in `/`.
8. Sign the APK/AAB with the release keystore held outside the repository (IMP-07). **Recommendation**
9. Do not seed test accounts on a public host (AUD-14). **Recommendation**
10. Add a dated entry to `PROJECT_IMPROVEMENTS.md`; tag the commit and create a GitHub release with the signed artifacts. **Recommendation**

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 08

Engineering Audit

Evidence-based findings with severity, confidence, affected area and recommended action, plus an addendum verifying earlier remediation.

Document control

Document ID	NX-08-AUD
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Development team prioritising fixes; assessor reviewing quality assurance
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-08-AUD contents

1. 1 Scope and method
2. 2 Summary
3. 3 Findings
4. 4 Addendum: verification of earlier remediation
5. 5 Addendum B: remediation in v2.0 (2026-11-05)

1 Scope and method

This audit covers the repository at the evidence baseline: Android source, backend source, build and CI configuration, tests, and existing documentation. The methods were code reading, running both test suites, retrieving GitHub Actions run history, and one targeted reproduction (AUD-01) in a scratch directory.

> **Not a security audit.** No penetration test, threat model, dependency vulnerability scan (`npm audit`/OWASP) or formal security review was performed. Security-relevant observations below come from reading code and must not be read as a security assessment.

1.1 Ratings

- **High**: likely data loss, broken core function, or blocked release. **Medium**: real user or operational impact under plausible conditions. **Low**: limited impact or quality debt. **Info**: observation.
- Confidence: **Verified** (reproduced or shown in logs/tests), **High** (unambiguous in code/config), **Medium** (depends on environment or library behaviour not exercised here).

2 Summary

Findings by severity

Severity	Count	IDs
High	3	AUD-01, AUD-03, AUD-04
Medium	6	AUD-02, AUD-05, AUD-06, AUD-08, AUD-18, AUD-19
Low	12	AUD-07, AUD-09, AUD-10, AUD-11, AUD-12, AUD-13, AUD-14, AUD-16, AUD-17, AUD-21, AUD-22, AUD-23
Info	2	AUD-15, AUD-20

Strengths observed: transactional offline outbox with idempotent server receipts; consistent per-account data scoping on both tiers; strict startup secret check; atomic file writes; standard error shape; injected fakes that make ViewModels testable; honest scoping in the existing docs (the addendum explicitly forbids claiming unbuilt features).

3 Findings

Engineering findings

ID	Finding and evidence	Severity / confidence	Area	Recommended action
AUD-01	**Unreadable data file is silently replaced with an empty store.** `loadDatabase()` catches a JSON parse error, logs it, and continues with empty maps; the next write overwrites the file. Reproduced 2026-11-05: a truncated file loaded as 0 users and was rewritten as `{"users":[],...}` by the first `saveDatabase()`.	**High** / Verified	Backend persistence	Exit non-zero on a parse failure; keep a `.bak` before each rename. IMP-03

ID	Finding and evidence	Severity / confidence	Area	Recommended action
AUD-03	**Open Dependabot PR #27 (Zod 4) breaks all request validation.** `validate()` reads `result.error.errors`, which is undefined on Zod 4. CI run 2026-10-31 shows `TypeError … reading 'map'` at `server.js:133`, so invalid input would return 500 instead of 400. `z.record(z.unknown())` in `syncSchema` also uses the single-argument form that Zod 4 changed (not verified).	**High** (if merged) / Verified	Backend dependencies	Do not merge as-is; migrate to `error.issues` and `z.record(z.string(), z.unknown())`. IMP-02
AUD-04	**The current code has never passed CI.** Every workflow run for PRs #25 and #26 and their merges failed in 3–5 s with the annotation "job was not started because your account is locked due to a billing issue". The last green Android CI on main was 2026-09-24 (CI-RUN-GREEN).	**High** / Verified	Delivery	Resolve the account lock and re-run CI on main. IMP-01
AUD-02	**Root `docker-compose.yml` conflicts with the Docker image.** It sets `DATABASE_PATH=/usr/src/app/nexus-data.json` and bind-mounts a single host file there. The image runs as `node` with `WORKDIR /app`, so the server cannot create `nexus-data.json.tmp` in the root-owned `/usr/src/app`, and renaming over a bind-mounted file is not possible. If the host file does not exist, Docker creates a directory in its place. `backend/compose.yaml` (named volume at `/data`) is consistent.	**Medium** / Medium (not executed; Docker daemon unavailable)	Deployment config	Delete the root file or align it with `backend/compose.yaml`. IMP-05
AUD-05	**Cached user content is included in Android backups.** `allowBackup="true"`; the backup and extraction rules exclude only `sharedpref`, so `nexus_database` (projects, tasks, habits, pending queue) goes to cloud backup and device transfer.	**Medium** / High	Android privacy	Exclude the `database` domain, or document the decision. IMP-09
AUD-06	**Auth rate limiting will misbehave behind a TLS proxy.** `trust proxy` is not set, so `req.ip` is the proxy address and all users would share one 10-per-15-min bucket. Hosting requires a TLS proxy (OPS-01).	**Medium** / Medium	Backend security / ops	Configure `trust proxy` for the chosen host. IMP-06
AUD-08	**No release signing configuration.** `app/build.gradle.kts` defines no `signingConfig`, so CI release artifacts would be unsigned and could not be installed or published.	**Medium** / High	Release	Add signing from CI secrets. IMP-07

ID	Finding and evidence	Severity / confidence	Area	Recommended action
AUD-18	**Misleading sign-in and registration errors.** `toApiError()` maps every 401/403 to `SessionExpired`, so a wrong password shows "Your session has expired. Please log in again." A 409 (email taken) and a 429 (rate-limited) both show "Something went wrong. Please try again."	**Medium** / High	Android auth UX	Map auth-endpoint statuses separately. IMP-04
AUD-19	**One rejected operation blocks the whole upload queue.** Uploads stop at the first failure; a 400 or 409 needs the user to choose "Discard item edits". Until then no later change for any item uploads, though they stay saved locally.	**Medium** / High	Sync design	Keep the ordering guarantee per entity only, or surface the blocked item by name in Review. Document it in the user manual (done in NX-10-UM).
AUD-07	**Throughput is bounded by one global lock.** Every authenticated handler, including bcrypt cost-12 hashing and full-file rewrites, runs under `dbMutex`. Sync receipts (`syncOperations`) grow without limit, and dashboard refresh makes one request per project.	**Low** / High	Backend performance	Hash outside the lock; cap or expire receipts; add a bulk tasks endpoint. Revisit with the SQLite plan.
AUD-09	**Documentation drift.** README contents links to five sections that do not exist (Roadmap, Continuous Integration, Part 2 PoE Evidence, Security and Release Notes, Contributing). It claims "MyMemory translation fallback", but `OnlineLanguageRepository` has no callers. It labels a test user "Global system administrator" though no roles exist. It lists full multi-language support, which is partial. The project list header shows a stray "?" (encoding).	**Low** / High	Documentation	IMP-16
AUD-10	**"Task due-date reminders" switch has no effect.** No notification permission, channel, WorkManager or AlarmManager code exists; the switch only stores a preference.	**Low** / High	Android feature	Hide it, or implement reminders (OQ-01, IMP-08).
AUD-11	**CSV export is not escaped.** Rows are `"${name},${frequency},$date"`; commas, quotes or newlines in a habit name corrupt the file.	**Low** / High	Android export	IMP-13
AUD-12	**Room migrations are untested and incomplete.** `exportSchema = false`; there are no migrations for 1→2 or 2→3, and no destructive fallback, so a device with a v1/v2 database would fail to open it. There are no migration tests.	**Low** (no public release yet) / High	Android data	IMP-15

ID	Finding and evidence	Severity / confidence	Area	Recommended action
AUD-13	**Lint is configured but not enforced.** ktlint, detekt and Kover are in the build, but CI runs only `test` / `assemble`. The committed `build-verification.log` (UTF-16) records a failed local `ktlintFormat` ("KtLint failed to parse file … app/build.gradle.kts"). `blank.yml` is a starter template.	**Low** / High	Code quality / CI	IMP-10
AUD-14	**Test account passwords are committed.** `docs/seed_test_users.ps1` and the README contain plain-text passwords for three seed accounts. This is acceptable for local development, but the accounts must never be seeded on a hosted API.	**Low** / High	Security hygiene	Release checklist item REL-09; consider generating the passwords at seed time.
AUD-16	**Minor auth weaknesses.** Login returns before any bcrypt work for unknown emails, so timing could reveal registered emails. A legacy path accepts a stored non-bcrypt value as a plain-text password. Email uniqueness is case-sensitive.	**Low** / High	Backend auth	Compare against a dummy hash; remove the legacy path once no such records exist; normalise emails to lower case.
AUD-17	**Habit completion rate is understated with several habits.** `HabitScoring` counts distinct dates across all habits but sums the schedule per habit. Two daily habits both done for 30 days score 50%. Streaks also count a day if *any* habit was done.	**Low** / High	Domain logic	IMP-14
AUD-21	**UI screenshot set does not show the screens it names.** `docs/screenshots/ui-audit/` holds 9 files but only 4 distinct images (MD5): `dashboard-offline`, `profile`, `project-list` and `settings` are identical. `habits`, `project-list-skeleton`, `login` and `register` are also skeleton states, and `splash.png` shows the Login screen.	**Low** / Verified	Evidence / documentation	Recapture before submission; do not present these files as evidence of named screens. IMP-16
AUD-22	**Two instrumented tests depend on the API being unreachable.** `SplashIntroScreenTest` plants a fake token. When a live API is reachable at 10.0.2.2:5263 it answers 401, the dashboard signs out, and the "Log Out"/"Retry" assertions fail. Observed 2026-11-05: 14/16 with the API running, 16/16 with it stopped. (Found during v2.0 verification.)	**Low** / Verified	Test reliability	Inject fake repositories into `NexusNavHost` for UI tests instead of relying on network state.
AUD-23	**detekt cannot run.** `./gradlew :app:detekt` fails with an internal error whose message is only "25.0.3". This is consistent with detekt 1.23.8 not supporting the current Kotlin/JDK toolchain. (Found during v2.0 lint work.)	**Low** / Verified	Code quality tooling	Upgrade to detekt 2.x when it supports this toolchain, or remove the plugin.

ID	Finding and evidence	Severity / confidence	Area	Recommended action
AUD-15	**Dead code.** `AuthSession.OFFLINE_TOKEN_PREFIX` handling remains after synthetic offline auth was removed; `setOnSessionCleared` is never registered; `OnlineLanguageRepository` and `languageInstance` are unused.	**Info** / High	Android	Remove during the next refactor.
AUD-20	**Inert or ordering quirks in the API.** `registrationEnabled` in `/app/config` is not enforced. Validation runs before authentication on several routes, so an unauthenticated request with a bad body gets 400 instead of 401. CORS is `*` (documented as intentional).	**Info** / High	Backend	Address when roles or a web client are introduced.

4 Addendum: verification of earlier remediation

`PROJECT_IMPROVEMENTS.md` records two remediation passes (2026-09-23 backend, 2026-09-24 Android). Each claim was checked against the baseline code and test evidence.

Remediation claims versus evidence

Claimed remediation	Code evidence	Test evidence (2026-11-05)	Result
AsyncMutex + atomic `.tmp` / `rename` writes	`AsyncMutex`, `withDb`, `saveDatabase()`	"Concurrent PATCH requests …" passed	Confirmed; see AUD-01 for the remaining load-path gap
Zod validation on all write endpoints	Schemas for register, login, profile, password, project, task, habit, sync	Validation tests passed (Zod 3)	Confirmed on Zod 3; at risk with Zod 4 (AUD-03)
JWT secret ≥ 32 chars, 24 h expiry, bcrypt 12, 10/15 min rate limit	`server.js` lines for each	Startup-refusal, expired-token and 429 tests passed	Confirmed; proxy caveat AUD-06
`/health` + Pino with redaction	Routes and `redact.paths`	Health test passed	Confirmed
Repository split, manual DI, `HabitScoring`, type-safe routes, shared fakes	`data/*Repository.kt`, `NexusApp`, `domain/`, `Route`, `test/fakes/`	32/32 unit tests passed	Confirmed
Synthetic offline authentication removed	No code mints offline tokens	Not separately tested	Confirmed, with leftover parsing code (AUD-15)

5 Addendum B: remediation in v2.0 (2026-11-05)

Part 3 fixed the findings below. Each fix was verified as stated; "not executed" means it was checked by reading only.

Finding status after Part 3

Finding	Status	Fix and verification

Finding	Status	Fix and verification
AUD-01	Fixed	Unreadable file: refuses to start, file untouched, `.bak` kept on every save. 3 Jest tests, including a real process start.
AUD-02	Fixed (config)	Root compose uses a named volume at `/data`; validated with `docker compose config`; not run.
AUD-03	Fixed	`error.issues`, two-argument `z.record`; Zod 4.6.7 installed; 29/29 tests.
AUD-04	Open (team)	Account billing lock; workflows updated to Node 22.
AUD-05	Fixed	`database` domain excluded from backup and transfer.
AUD-06	Fixed	`TRUST_PROXY` setting; not tested.
AUD-08	Fixed	Signing config from `keystore.properties` or env; upload key created; signed APK verified with `apksigner` (see NX-07-CICD).
AUD-09	Mostly fixed	README rewritten; stray "?" replaced; the admin label removed. The unused MyMemory client remains (AUD-15).
AUD-10	Fixed	Reminders implemented (FR-23); verified on device.
AUD-11	Fixed	RFC 4180 quoting; `CsvRowTest`.
AUD-16	Mostly fixed	Dummy bcrypt compare for unknown emails; emails lower-cased. Legacy plain-text path kept for old records.
AUD-17	Fixed	Per-habit completions; new unit test.
AUD-18	Fixed	Status-specific auth messages; `AuthErrorMessagesTest`; verified on device.
AUD-13	Fixed (ktlint)	0 ktlint violations; `ktlintCheck` gated in Android CI; placeholder workflow and failing log removed. detekt: AUD-23.
AUD-15	Fixed	Unused MyMemory client (`LanguageApiService`, `OnlineLanguageRepository`, `languageInstance`) and the offline-token parsing removed.
AUD-21	Fixed	20 new, correctly named screenshots in `docs/screenshots/part3/`.
New	Fixed	Settings crashed with "No ActivityResultRegistryOwner" after adding the permission request: `MainActivity` overrides `LocalContext`. Fixed by providing `LocalActivityResultRegistryOwner`; verified on device (crash buffer empty).

Still open: AUD-04, AUD-07, AUD-12, AUD-14, AUD-19, AUD-20, AUD-22, AUD-23. Part 3 added the SSO and push endpoints; like the rest of the API, they were reviewed by reading code and testing, not by a security audit.

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 09

Testing, Deployment and Handover

Test strategy and verified results, deployment summary, operational handover, known limitations and runbooks.

Document control

Document ID	NX-09-TDH
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Incoming maintainers and operators; assessor checking test evidence
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-09-TDH contents

1. 1 Test strategy
2. 2 Verified results
3. 3 Deployment summary
4. 4 Operational handover
5. 5 Known limitations
6. 6 Runbooks

1 Test strategy

Test levels in the repository

Level	Tooling	What it covers	Runs in CI
Backend unit/integration	Jest + Supertest against the in-process Express app	Validation, auth, rate limiting, error shape, health, body limit, concurrency, sync replay, ownership and conflicts	Yes (CI-01, CI-02)
Android unit (JVM)	JUnit 4, MockK, coroutines-test, shared fakes	ViewModels (auth, dashboard, projects, settings), `SyncManager`, `HabitScoring`	Yes (`./gradlew test`)
Android instrumented	AndroidX JUnit, Compose UI test	Room offline persistence and ownership, splash with reduced motion, screenshot-capture wait conditions	No; needs an emulator or device
Container smoke	Docker + curl `/health`	Image builds and starts in production mode	Backend CI only; never succeeded (CI-02)
Manual acceptance	`docs/ONLINE_OFFLINE.md` six-step check; `docs/DEMO_SCRIPT.md`	Online/offline round trip, account switch, conflicts	No. No recorded results in the repository **Unknown**

2 Verified results

> **v2.0 results (2026-11-05, after the Part 3 changes):** backend **29/29** (adds `part3.test.js`, 13 tests: SSO, push, device tokens, data-file safety, email case). Android unit **42/42** (adds `AuthErrorMessagesTest` 5, `ReminderPlannerTest` 2, `CsvRowTest` 2, one `HabitScoringTest` case). Instrumented **16/16** on an Android 7.0 emulator with the API stopped (AUD-22 explains the 14/16 run with a live API). The on-device walkthrough covered the sign-in error, all screens, three languages, the offline round-trip, the reminder and test notifications, and the Settings crash fix. The tables below are the v1.0 baseline results.

v1.0 run on 2026-11-05 against commit 3a3f275, before the Part 3 changes.

Backend: `cd backend; npm test`, 2 suites, 16 tests, 16 passed, 31.7 s

ID	Suite	Tests	Result
TST-01	`tests/server.test.js`: JWT secret refusal; concurrent updates; Zod validation (register, task enum/date, strict profile); error shape and production masking; expired token 401; 429 after 10 logins; `/health`; 413 over 1 MB	11	Passed
TST-02	`tests/sync.test.js`: ordered replay with cascade delete; lost response plus restart does not duplicate or resurrect; cross-account project rejected; invalid payload 400 and no-auth 401; edit of deleted item gives 409	5	Passed

Android: `./gradlew :app:testDebugUnitTest --rerun`, 7 classes, 32 tests, 0 failures, BUILD SUCCESSFUL

ID	Class	Tests	Result
TST-03	`AuthViewModelTest`	9	Passed
TST-04	`DashboardViewModelTest`	7	Passed

ID	Class	Tests	Result
TST-05	`SyncManagerTest` : acknowledged operations removed in sequence; network failure keeps pending changes; pending writes block refresh overwrite; offline reads use cache while auth errors surface	4	Passed
TST-06	`HabitScoringTest`	4	Passed
TST-07	`SettingsViewModelTest`	4	Passed
TST-08	`ProjectListViewModelTest`	3	Passed
TST-09	`ExampleUnitTest` (template)	1	Passed

Not executed for this package

ID	Test	Count	Status
TST-10	Instrumented: `OfflinePersistenceTest` (1), `SplashIntroScreenTest` (4), `ScreenshotCaptureTest` (10), `ExampleInstrumentedTest` (1)	16	Not run at v1.0; run at v2.0: 16/16 passed on an emulator
TST-11	Docker build and smoke test	1	Not run; Docker daemon not running Unknown
TST-12	Manual acceptance check (`ONLINE_OFFLINE.md`)	6 steps	v2.0: steps 1–4 exercised on the emulator (offline create, then reconnect, then present on the server, no duplicate); steps 5–6 (account switch, cross-client delete conflict) not run Unknown

2.1 Coverage gaps

- No unit tests for `ProjectDetailViewModel` , `TaskListViewModel` , `HabitsViewModel` , `RemoteConfigViewModel` , or the repositories' Room paths Implemented .
- No backend tests for habits CRUD, password change, account deletion or the localisation/config endpoints Implemented .
- No Room migration tests (AUD-12); no tests for the defects in AUD-01, AUD-11, AUD-17 and AUD-18.
- A Kover report is configured but no coverage figure was produced or recorded Unknown .

3 Deployment summary

Full procedures are in NX-11-DEP; release gating is in REL-CHECK. In short: run the API with a persistent `JWT_SECRET` and persistent storage (Node host or `backend/compose.yaml`), put it behind HTTPS, and build the app with `-PNEXUS_API_URL=https://…/api/` . No environment beyond local development exists today (CON-02).

4 Operational handover

Handover register

Item	Current state
Source repository	`github.com/jBlack-MC/Nexus-app` ; default branch `main` ; working branch `New-Changes` Implemented
Repository owner / admins	Owner account `jBlack-MC` ; other collaborators Unknown

Item	Current state
Secrets in use	`JWT_SECRET`; since v2.0 also `GOOGLE_CLIENT_ID` (not secret), `FIREBASE_SERVICE_ACCOUNT_BASE64` (secret), and the upload keystore at `%USERPROFILE%\.nexus\nexus-upload.jks` with its password in git-ignored `keystore.properties` on the development PC. **Back these up.** CI uses a hard-coded test-only JWT value Implemented
CI variables	`NEXUS_API_URL` (enables release artifacts). Whether it is set: Unknown. Release jobs were skipped in the retrieved runs Inferred
Hosted services	None Implemented
Monitoring and alerting	`/health` endpoint and Docker `HEALTHCHECK`. No external monitoring Implemented
Backups	None configured Implemented. See RB-04
Open pull requests	#27 Dependabot Zod 4. Do not merge until IMP-02 is done

5 Known limitations

- Single-instance JSON store (CON-01); not suited to concurrent hosting or large data.
- No background sync while the app process is stopped (CON-03).
- Last-upload-wins for edits; no field-level merge (FR-17).
- Account recovery and collaboration are not implemented (NX-01-PRD scope). Google SSO and FCM push need the team's Google Cloud and Firebase projects before they work end to end.
- isiZulu and Setswana translations are AI-drafted and need first-language review (FR-18).
- Open defects: see NX-08-AUD.

6 Runbooks

Operational runbooks

ID	Situation	Steps
RB-01	API exits at start with "FATAL: JWT_SECRET is missing, too short..."	Generate a secret: `node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"`. Set `JWT_SECRET` in the host's secret store (not in git) and restart. Reuse the same value on later restarts.
RB-02	Health check	`curl -s https://<host>/health` should return `{"status":"ok","uptime":…}`. Locally: `http://localhost:5263/health`.
RB-03	All users suddenly see empty data after a restart	Likely AUD-01. **Stop the API immediately** so no further writes occur. Inspect logs for "Failed to load database". Restore the data file from the latest backup (RB-04) and restart. Without a backup, the data overwritten by the empty store cannot be recovered.
RB-04	Back up and restore data Recommendation	Back up: stop the API, or accept a crash-consistent copy (writes are atomic renames), then copy `nexus-data.json` (container: `docker compose cp api:/data/nexus-data.json ./backup-YYYYMMDD.json`). Restore: stop the API, replace the file, start the API, check RB-02. Treat backups as personal data.

ID	Situation	Steps
RB-05	Users cannot sign in (HTTP 429)	The limit is 10 auth requests per 15 minutes per IP, held in memory. Wait 15 minutes or restart the API. Behind a proxy, check AUD-06.
RB-06	Rotate `JWT_SECRET`	Set the new value and restart. Every existing session becomes invalid; users see "Sign in again to sync saved changes", and their queued changes upload after they sign in.
RB-07	A user reports the sync banner stuck on "review required"	Ask them to tap Review. "Discard item edits" removes only that item's queued edits, then the rest uploads. If they choose "Keep changes", nothing uploads until the conflict is resolved (AUD-19).
RB-08	CI runs fail within seconds	Open the run with `gh run view <id>` and read the annotations. If they show "job was not started", the cause is an account or billing restriction, not the code (AUD-04).

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 10

User Manual

Task-based instructions for Nexus users: account, projects, tasks, habits, settings and working offline.

Document control

Document ID	NX-10-UM
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	People using the Nexus Android app (signed-in users) and testers following the demo
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-10-UM contents

1. 1 Before you start
2. 2 Account
3. 3 Today screen
4. 4 Projects and tasks
5. 5 Habits
6. 6 Settings
7. 7 Working offline

1 Before you start

- You need an Android phone or emulator running Android 7.0 or later, with the Nexus app installed from a build provided by the team. No app-store listing exists.
- Your first registration or sign-in must happen while the device can reach the Nexus server. After that you can keep working offline.
- Nexus has one kind of user. Everything you create is private to your account, and there are no shared projects.

> Screen labels in this manual are the English ones. Choose isiZulu or Setswana on the Login screen or in Settings → Language, and every screen, message and notification switches language.

2 Account

2.1 Create an account (UM-01)

1. Open Nexus and wait for the intro animation to finish. The Login screen opens.
2. Optional: choose your language in the **Language** box.
3. Tap **Need an account? Register**.
4. Enter **Display Name**, **Email** and a **Password** of at least 6 characters, then tap **Create Account**.
5. The Today screen opens. You stay signed in when you close and reopen the app (for up to 24 hours per sign-in).

If the email is already registered, the app says so. Sign in instead.

2.2 Sign in and sign out (UM-02)

1. On Login, enter your Email and Password and tap **Sign In**.
2. To sign out, tap **Log Out** on the Today screen or in Settings, then confirm.

If the email or password is wrong, the app says "Email or password is incorrect." After 10 attempts in 15 minutes it asks you to wait.

2.3 Continue with Google (UM-12)

1. On Login or Register, tap **Continue with Google**. This button only appears in builds configured for Google sign-in.
2. Choose your Google account. Your Nexus account is created on first use, or linked if you already registered with the same email.
3. Accounts created with Google have no Nexus password; keep using **Continue with Google**.

Changes you saved while offline stay on the device after you sign out. They upload the next time *you* sign in on that device. Another person signing in on the same phone will not see them.

3 Today screen

The **Today** tab shows how many projects and tasks you have and how many tasks you've completed ("Activity"). Under **Your schedule** are three cards:

- **Overdue**: unfinished tasks with a due date before today.
- **Due today**: unfinished tasks due today.

- **Upcoming**: the next three unfinished tasks with a future due date.

Tasks without a due date do not appear in these cards. "Showing saved data" means the numbers come from your device because the server could not be reached.

4 Projects and tasks

4.1 Create and manage a project (UM-03)

1. Tap the **Projects** tab, then the **+** button.
2. Enter a **Name** and an optional description, then confirm.
3. Use **Search projects** to filter the list by name.
4. Tap a project to open it. There you can rename it (**Save Project**), add tasks, open the task list (**Open Tasks Screen**) or **Delete Project**.

Deleting a project also deletes all its tasks. Nexus asks you to confirm every delete; there is no undo afterwards.

4.2 Add and edit tasks (UM-04)

1. In a project's task list, tap **+** (New Task).
2. Enter a **Title**. Optionally add a description, a **Due date** (calendar picker; the × clears it), **Priority** (None, Low, Medium, High) and **Status** (To do, In progress, Done).
3. To add labels, type in **Add label** and tap **Add**; tap × on a label to remove it. Add checklist steps the same way with **Add step**.
4. Save. Tap a task to see its details, then **Edit task** to change it. Setting the status to Done marks the task complete.

4.3 Find tasks (UM-05)

- **Search tasks or labels** matches titles and labels.
- Status chips: **All**, **Open**, **In progress**, **Done**. Tap **High priority** to show only high-priority tasks.
- The bar at the top shows "x of y tasks complete".

5 Habits

5.1 Track a habit (UM-06)

1. Tap the **Habits** tab, then **+** (Add habit).
2. Enter a **Habit name**, choose **Daily**, **Weekly** or **Custom**, and tap **Create**.
3. Each day you do the habit, tap **Complete today**. Tap it again (it reads **Completed today**) to undo.
4. Use the pencil icon to edit, or the delete icon to remove a habit.

The stats card shows your level, points, current and best streak (in days), completion rate and badges ("3-day streak", "7-day streak", "Reliable habit builder"). The completion rate counts each habit against its own schedule.

6 Settings

6.1 Change your profile and preferences (UM-07)

- **Display Name**: edit it in the User Profile section and tap **Save Changes**. Your email cannot be changed.
- **Appearance**: System, Light or Dark. This applies to this device only.

- **Language**: English, isiZulu or Setswana. It is saved to your account.
- **Task due-date reminders**: when on, Nexus notifies you about overdue and due-today tasks, about twice a day, even offline. On Android 13 and later, allow notifications when asked. You also get an alert when your account changes on another of your devices.
- **Send test notification**: shows a notification on this phone and reports whether the server could push to your devices.
- **Sync account** reloads your profile from the server.

6.2 Export your habit history (UM-08)

1. In Settings, tap **Export habits CSV**, then **Share CSV**.
2. Choose an app (email, Drive, etc.). The file lists habit, frequency and each completed date. Habit names with commas or quotes are exported correctly.

6.3 Change password or delete your account (UM-09)

- **Change password**: enter your current password and a new one (6+ characters), then tap Save. You must be online.
- **Delete account and data**: confirm **Delete**. Your account and everything in it are permanently removed from the server and this device. This cannot be undone. You must be online.

7 Working offline

7.1 Understand the sync banner (UM-10)

The banner at the top of every screen tells you where your changes are:

Sync banner messages

Message	Meaning	What to do
Saved on device · waiting to sync (*n* pending changes)	Your change is safe on the phone and not yet on the server.	Nothing. It uploads automatically when you're online, or tap **Sync**.
All changes synced	The server has everything.	Nothing.
Offline · changes saved on device / Offline · showing saved data	The server can't be reached.	Keep working; reconnect later.
Sign in again to sync saved changes	Your session expired.	Tap **Sign in**. Your saved changes are kept.
An item changed on another device · review required	The server rejected one change, for example because you edited an item that was deleted elsewhere.	See UM-11.

Uploads happen while Nexus is open. If Android closes the app, uploading resumes when you open it again. Items you never opened or refreshed while online won't be available offline.

7.2 Resolve a conflict (UM-11)

1. Tap **Review** on the banner.
2. Choose **Discard item edits** to drop your unsent changes to that one item. Your other changes then upload. Or choose **Keep changes** to decide later.

3. Refresh the screen to load the server's current copy.

Until you discard the conflicting item, none of your other pending changes upload (AUD-19). They stay safely on your device.

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 11

Deployment Guide

How the API and app are actually deployed today (local), the container path the repository supports, and labelled recommendations for public hosting.

Document control

Document ID	NX-11-DEP
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Developers and operators setting up the API and building the app
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-11-DEP contents

1. 1 Current deployment position
2. 2 Prerequisites
3. 3 Run the API locally (DEP-01)
4. 4 Run the API in a container (DEP-02)
5. 5 Public hosting (recommendations)
6. 6 Rollback
7. 7 Enabling Google SSO, push and signing (v2.0)

1 Current deployment position

> **No hosting platform is configured.** The repository contains no provider-specific configuration: no cloud manifests, infrastructure-as-code or deploy workflow. The README states that public deployment "still needs a hosted API URL" and that the earlier production hostname "was not a configured service" Implemented. Sections 3 and 4 document what the repository supports, and section 5 contains recommendations only.

Supported deployment targets

ID	Target	Defined by	Verified
DEP-01	Local Node process (development)	`backend/package.json` `start` / `dev`, `docs/ONLINE_OFFLINE.md`	Server module loaded and tested in process (16/16) Implemented; listening server not started for this package
DEP-02	Single container with a named volume	`backend/Dockerfile`, `backend/compose.yaml`	Not executed Unknown; CI smoke test never succeeded (CI-02)
DEP-03	Root `docker-compose.yml`	Repository root	Fixed in v2.0 to match DEP-02 (named volume at `/data`); config validated, not run (AUD-02)
DEP-04	Android debug/release builds	`app/build.gradle.kts`, Android CI	Debug build last green in CI on 2026-09-24; release never built in the retrieved history

2 Prerequisites

- API: Node.js 22 or later (required by google-auth-library 11 and firebase-admin 14); npm. For containers: Docker with Compose v2.
- App: Android Studio with SDK Platform 37; JDK 17+ (the build uses a 17 toolchain); an emulator or device on API 24+.
- A strong `JWT_SECRET` of 32+ characters, stored outside git (`.env*` files are git-ignored).

3 Run the API locally (DEP-01)

```
# PowerShell, from backend/
npm.cmd ci
$env:JWT_SECRET = node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"
npm.cmd start          # listens on http://localhost:5263
curl.exe http://localhost:5263/health
```

- Data is written to `backend/nexus-data.json` unless `DATABASE_PATH` is set. The file is git-ignored.
- Keep the same secret across restarts, or every session becomes invalid (RB-06).
- Optional test accounts: run `docs/seed_test_users.ps1` against a **local** API only (AUD-14).

3.1 Point the app at the local API

Debug build targets

Device	Command

Device	Command
Emulator	Run from Android Studio, or `.\gradlew.bat :app:assembleDebug`. The default URL `http://10.0.2.2:5263/api/` reaches the host.
Phone on the same LAN	`.\gradlew.bat :app:assembleDebug -PNEXUS_API_URL=http://<your-LAN-IP>:5263/api/`, and allow port 5263 through the firewall.

The URL must end in `/` and match `https?://host[:port]/path/`, or the Gradle build fails at configuration time (TR-07).

4 Run the API in a container (DEP-02)

```
# from backend/, with JWT_SECRET exported in the shell or a git-ignored .env file
docker compose up --build -d
docker compose ps          # STATUS should become "healthy"
curl http://localhost:5263/health
```

- The image is `node:22-alpine`. It installs production dependencies only, runs as the non-root `node` user, exposes 5263, and stores data at `/data/nexus-data.json` on the `nexus-data` named volume.
- A Docker `HEALTHCHECK` calls `/health` every 30 s. `restart: unless-stopped` is set.
- Compose refuses to start if `JWT_SECRET` is unset (`${JWT_SECRET:?…}`).
- Run **one** replica only (CON-01). Back up the volume (RB-04).

5 Public hosting (recommendations)

Everything in this section is **Recommendation**. It describes the minimum the code requires from any host; it does not select a provider.

Host requirements derived from the code

ID	Requirement	Why
DEP-R1	Runs a Node 20+ process or an OCI container, with exactly one instance	In-memory state and single-file store (CON-01)
DEP-R2	Persistent disk or volume mounted at `/data` (or `DATABASE_PATH`) that survives redeploys	Ephemeral file systems would lose all data
DEP-R3	HTTPS termination with a valid certificate and a stable domain	Release builds reject HTTP (NFR-02)
DEP-R4	Secret store for `JWT_SECRET`; `NODE_ENV=production`	Startup check; production error masking
DEP-R5	Health check against `/health`	Liveness
DEP-R6	Configure Express `trust proxy` before go-live	Rate limiting behind the proxy (AUD-06, IMP-06)
DEP-R7	Fix AUD-01 and set up scheduled backups before real users	Data-loss risk

5.1 Building a release that uses the hosted API

1. Confirm `https://<host>/health` responds.

2. Locally: `.\gradlew.bat :app:assembleRelease -PNEXUS_API_URL=https://<host>/api/` ; or set the GitHub repository variable `NEXUS_API_URL` so CI builds the release APK and AAB (CI-01).
3. Sign the artifacts. Signing is not configured in the repository (AUD-08).
4. Complete the release checklist (REL-CHECK).

6 Rollback

No versioned deployments exist, so rollback is manual **Implemented**. **Recommendation** Tag each deployed commit. To roll back, redeploy the previous tag against the same data volume. Data written by a newer version stays in the file (the JSON store has no schema version), so take a backup before every deploy.

7 Enabling Google SSO, push and signing (v2.0)

One-off setup per environment (Recommendation: the accounts are the team's to create)

Feature	Steps
Google SSO	1. Google Cloud Console → APIs & Services → Credentials: create an OAuth client of type **Web application** (its ID is used by both app and API) and one of type **Android** for package `com.example.nexus` with the SHA-1 of the debug key and of the upload key (printed by `scripts/create-upload-keystore.ps1`). 2. API: `GOOGLE_CLIENT_ID=<web-id>` . 3. App: `-PGOOGLE_WEB_CLIENT_ID=<web-id>` . 4. Check: `GET /api/app/config` shows `googleSignIn: true` .
Push (FCM)	1. Firebase Console: add an Android app `com.example.nexus` , download `google-services.json` into `app/` (git-ignored). 2. Project settings → Service accounts → generate key; set `FIREBASE_SERVICE_ACCOUNT_BASE64` to the base64 of that JSON on the API host. 3. Check: `pushNotifications: true` in `/api/app/config` , then Settings → Send test notification reports "Push sent to n device(s)".
Release signing	Run `scripts/create-upload-keystore.ps1` once (done on the development PC on 2026-11-05); back up `%USERPROFILE%\.nexus\nexus-upload.jks` and `keystore.properties` . In CI, provide `NEXUS_KEYSTORE_PATH` , `NEXUS_KEYSTORE_PASSWORD` , `NEXUS_KEY_ALIAS` and `NEXUS_KEY_PASSWORD` from secrets.
Proxy	Behind an HTTPS reverse proxy set `TRUST_PROXY=1` (DEP-R6).

NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 12

Team Contribution Report

Verifiable contribution evidence from the repository and GitHub, with unavailable evidence marked as unknown.

Document control

Document ID	NX-12-TCR
Version	2.0
Date	2026-11-05
Status	Draft for team review (not approved)
Intended readers	Module assessor; team members confirming their contributions
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

NX-12-TCR contents

1. 1 Basis of this report
2. 2 Declared team
3. 3 Recorded activity by account
4. 4 Tool-assisted work
5. 5 Contribution areas
6. 6 Gaps and how to close them
7. 7 Part 3 (v2.0) work

1 Basis of this report

This report lists only what the repository and the GitHub API record. It does not judge the size or quality of anyone's contribution, and it does not infer contribution from commit counts. Work done outside GitHub (testing sessions, design, reviews in person, recording the demo) leaves no trace here. Its absence from this report is *not* evidence that it did not happen.

Sources queried on 2026-11-05: `git log --all`, `git shortlog`, `gh pr list`, PR reviews, `gh issue list`, the repository collaborators API, and the README team table.

2 Declared team

Team as declared in README.md (self-reported; student numbers are listed there)

Name	Declared role	Linked GitHub account
Clarity Masuku	Backend/API Integration Android App Developer / UI & Authentication	Unknown Not stated in the repository; the team should confirm
Sibusiso Mabena	Testing & Quality Assurance	Unknown Not stated in the repository; the team should confirm

Repository collaborators (GitHub API): `jBlack-MC` (owner) and `SibusisoMabena05` Implemented. The similarity between the second account name and a declared member is noted, but it is not treated as proof of identity.

3 Recorded activity by account

Activity recorded on GitHub and in git

Account	Commits (all refs)	Pull requests opened	PR reviews	Issues opened
`jBlack-MC`	97, including 25 automatic "cline checkpoint" entries and stash-style "index on"/"untracked files on" entries created by tooling	18 (all merged)	18 "commented" reviews, all on its own PRs	8 (#1–#8)
`SibusisoMabena05`	0	0	0	0
`copilot-swe-agent[bot]`	2 (PR #23: "Clarify Android CI workflow", merge of main)	None	None	None
`dependabot[bot]`	1	1 (#27, open)	None	None

No pull request has an approving review from a second person Implemented.

4 Tool-assisted work

Commit history shows AI-assisted tooling: Cline checkpoints, two commits by GitHub's Copilot coding agent, and Dependabot Implemented. This documentation package was itself prepared with Claude Code. Commits made

through an account do not show whether a person or a tool wrote the change, so authorship within `jBlack-MC` commits cannot be split further Unknown.

5 Contribution areas

Areas of the project and attributable evidence

Area	Evidence of who did the work
Android app (UI, auth, data, sync)	Commits and PRs from `jBlack-MC` Implemented; individual person Unknown
Backend API	Commits and PRs from `jBlack-MC` Implemented; individual person Unknown
Testing and QA	Test files were committed by `jBlack-MC` Implemented. Manual test sessions, test planning or QA sign-off by the declared QA member: Unknown (no records in the repository)
CI configuration	`jBlack-MC` commits; one clarification commit by the Copilot agent Implemented
Demo recording and Part 2 PDF	`docs/Nexus-Part-2-Submission.pdf` committed by `jBlack-MC` Implemented; who produced it Unknown

6 Gaps and how to close them

- Open question Each member confirms which GitHub account is theirs (if any), and the README is updated with that mapping.
- Recommendation Record non-code work in the repository: test-session notes with dates and results under `docs/`, issues assigned to the person doing the work, and PR reviews from the other member, which also adds a second-person review gate.
- Recommendation If both members commit from a shared machine, set per-person `git config user.name` / `user.email`, or use `Co-authored-by:` trailers.

Until these gaps are closed, individual contribution claims in the PoE should rest on the members' own statements, not on this report.

7 Part 3 (v2.0) work

On 2026-11-05 the Part 3 changes (SSO, push, reminders, localisation, fixes, Play kit, documentation updates) were made with Claude Code in a session run from the `jBlack-MC` workspace. Commit `716bedb` ("Add Nexus documentation package") includes part of that work; the rest was uncommitted when this report was generated. As before, the person directing the session cannot be identified from the repository Unknown, and the team should state it in the AI-use write-up.